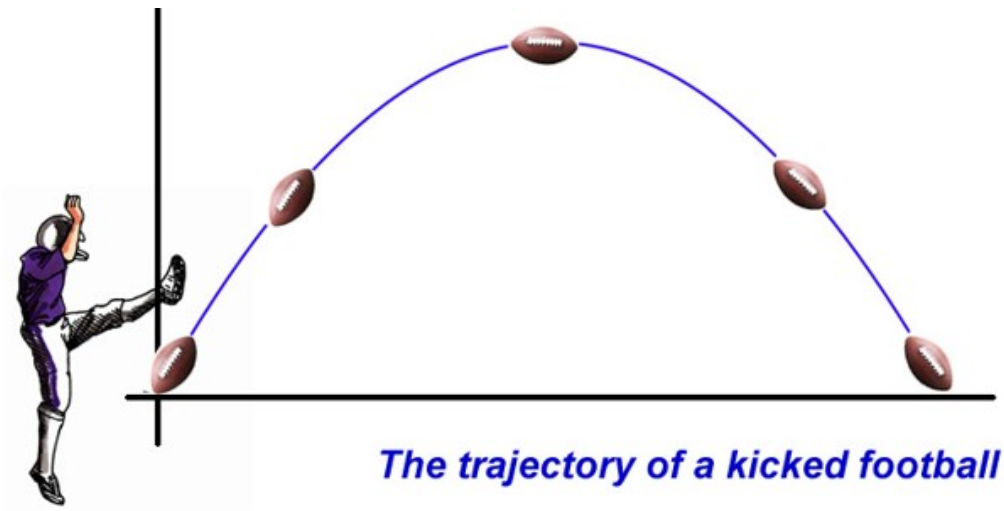
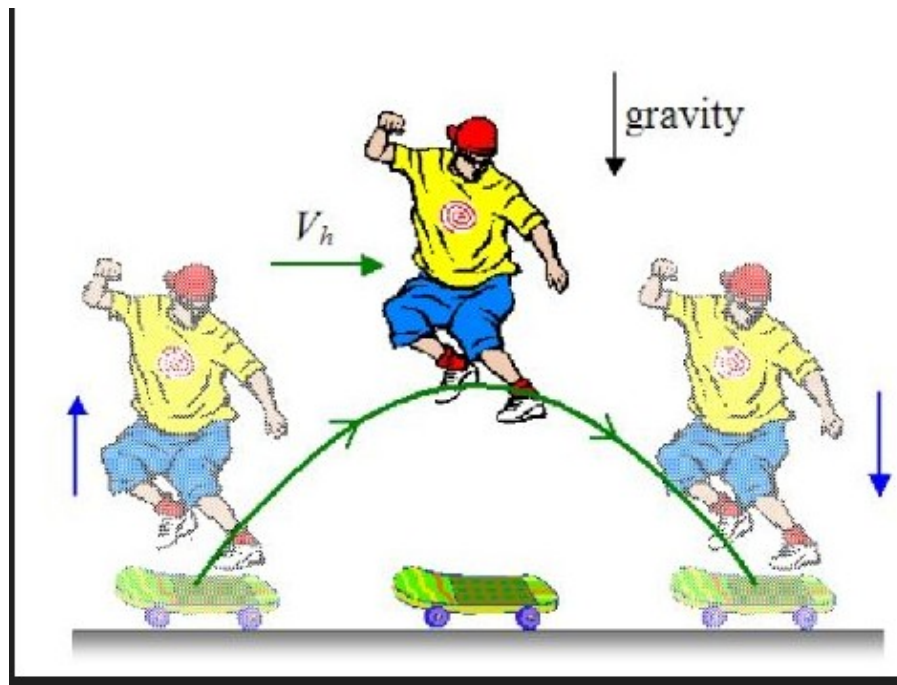


Vectors

Projectile motion
uniform circular motion



Hippie-Jump



Any 2 D motion can be decomposed
Into two independent motions:
One along the horizontal and one
Along the vertical/

→ we break down a 2D problem
To 2 1D

→ The x-motion takes place as if the
Y-motion isn't happening. The
Y motion takes place independent
Of whatever is happening in
The x direction

→ We get 2 sets of
Kinematics equation

What are theses equations
For the jumper ?

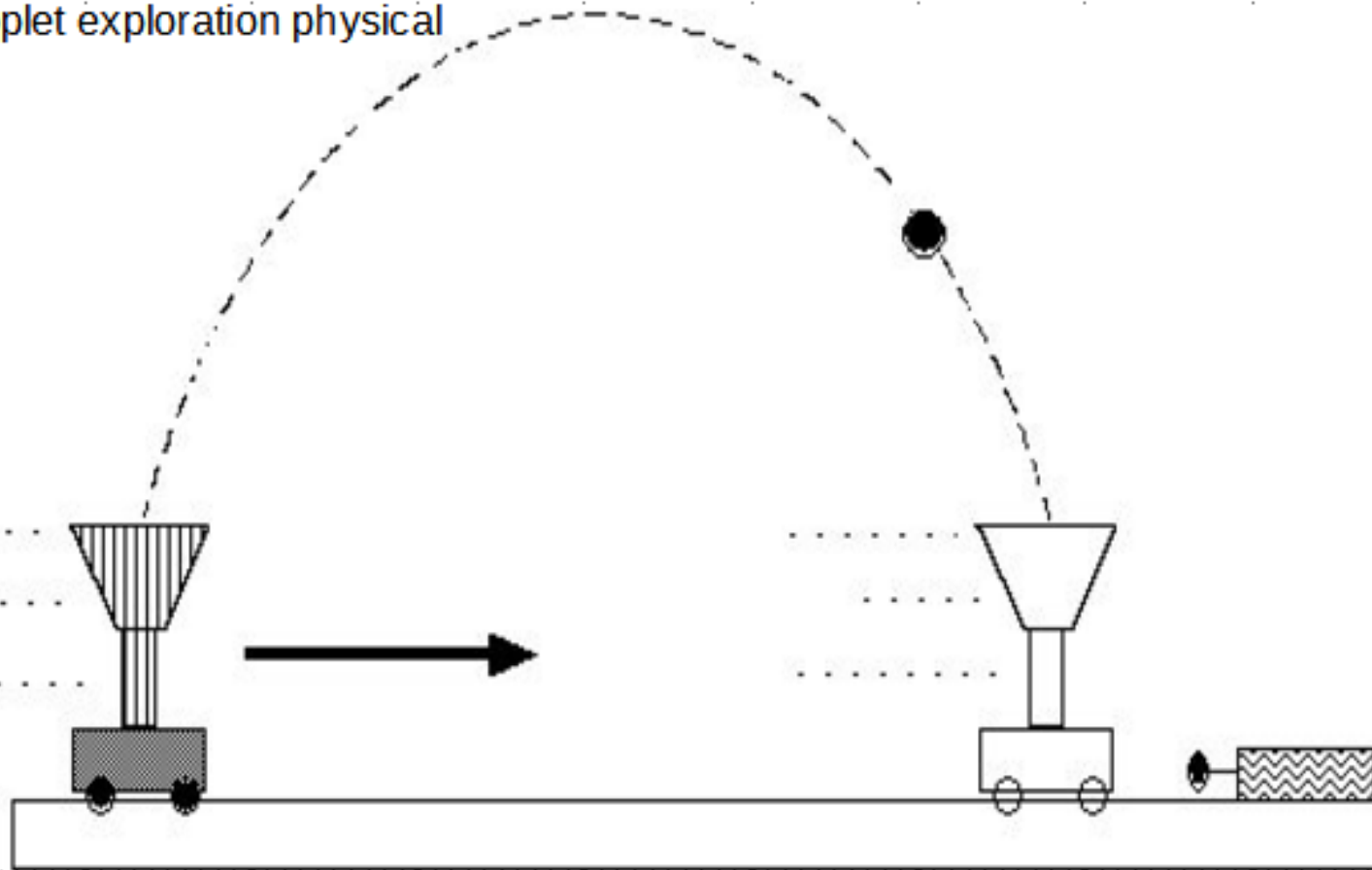
Videos
kine2D_projectile
Projectile trampoline
VTD05 video



Show Video

And applet exploration physical

Video



MOTION DEPENDS ON THE FRAME OF THE REFERENCE.

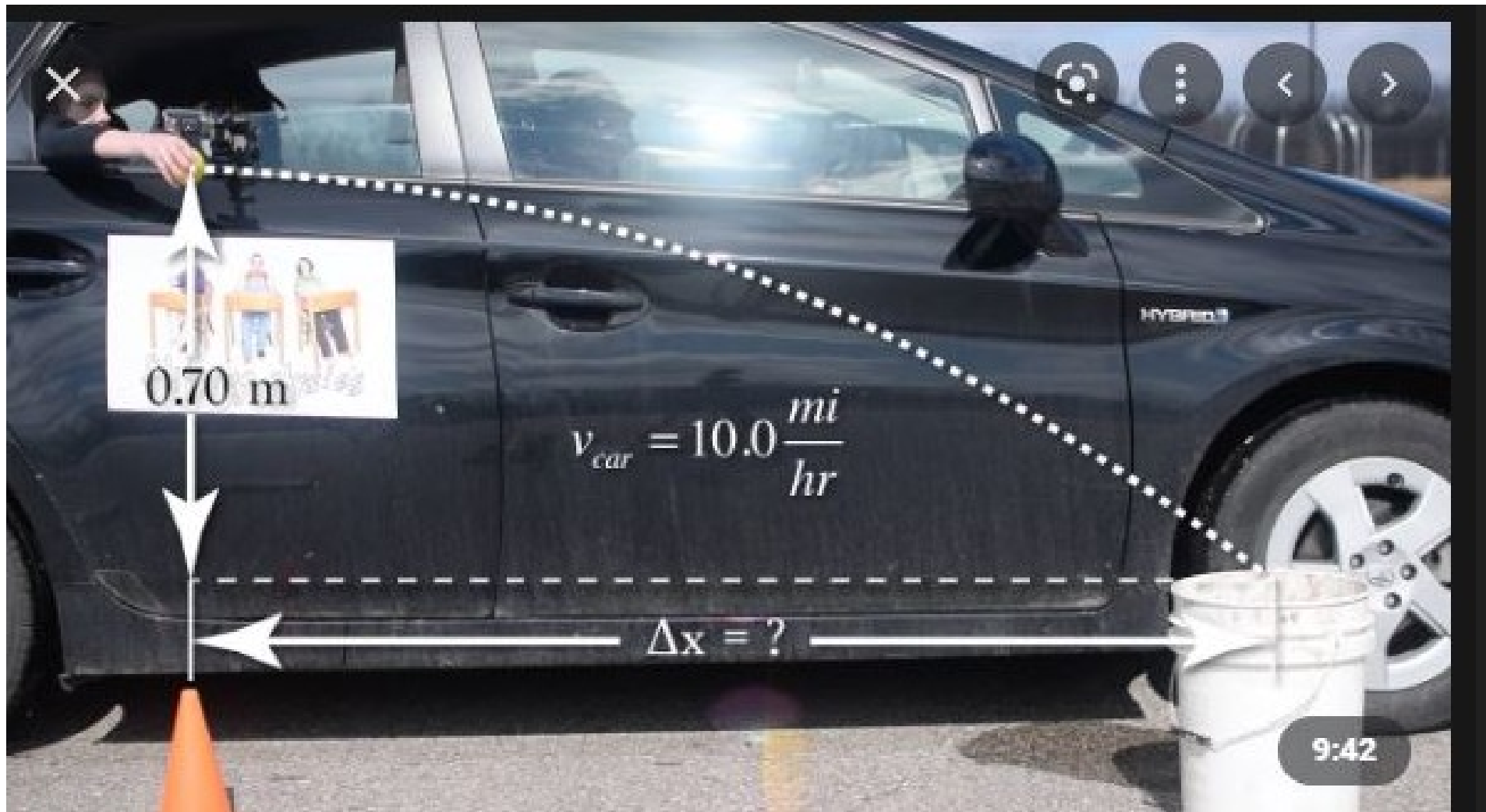
**If you stand on the car, the ball goes up and down along a straight line/
If you stand in the lab watching the car, the ball has a projectile motion.
It follows a parabola.**

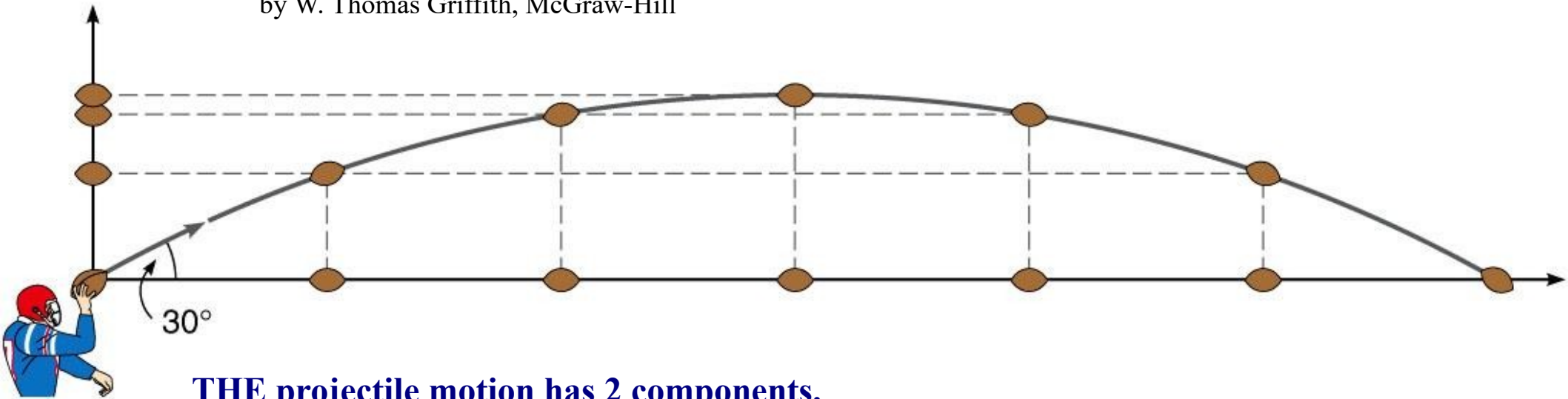
What are the 2 sets of equations here?

One set long the x-axis and one along the y-axis.

Say the speed of the car is 10m/s (its actually about 5m/s but to simplify)

How long to reach the ground? Distance covered?





THE projectile motion has 2 components.

Along the horizontal the acceleration is 0 (no force acting on the ball)

Along the vertical the acceleration = -10m/s/s

The horizontal motion ignores the vertical motion

See applet projectile motion - Use angle 70 degrees, initial position 0
observe the components of the velocity, the vector acceleration, force, the position

[http://www.walter-fendt.de/html5/phen/
projectile_en.htm](http://www.walter-fendt.de/html5/phen/projectile_en.htm)

https://phet.colorado.edu/sims/html/projectile-motion/latest/projectile-motion_en.html

Projectile motion

BIG IDEA

A two-dimensional problem
can be considered as 2
one-dimensional problems.

A body motion has 2 components
completely independent of each other.

2 components means : we will use trig to find the components of the velocities along the path.
A horizontal component (cosine if you use the standard notation) and vertical component
(sine if you use the standard notation).

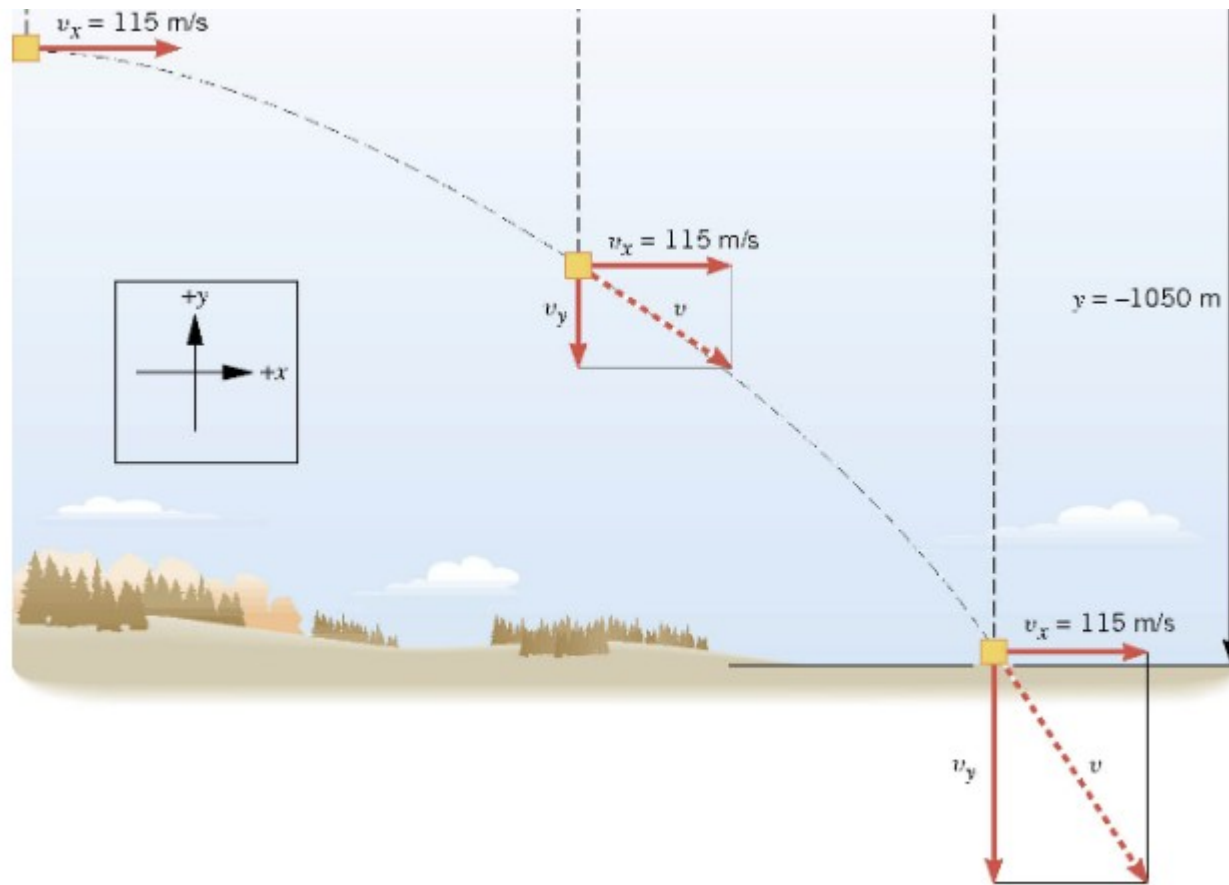
http://www.phy.hk/wiki/j/Eng/projectile/projectile_js.htm

A football is kicked at an angle 25° with respect to the horizontal. Which one of the following statements best describes the acceleration of the football during this event if air resistance is neglected?

- a) The acceleration is zero m/s^2 at all times.
- b) The acceleration is zero m/s^2 when the football has reached the highest point in its trajectory.
- c) The acceleration is positive as the football rises, and it is negative as the football falls.
- d) The acceleration starts at 9.8 m/s^2 and drops to some constant lower value as the ball approaches the ground.
- e) The acceleration is 9.8 m/s^2 at all times.

Example A simple projectile motion: initial velocity is horizontal
 $v_{0y} = 0$ and $v_{0x} = 115 \text{ m/s}$. The vertical displacement is -1050 m .

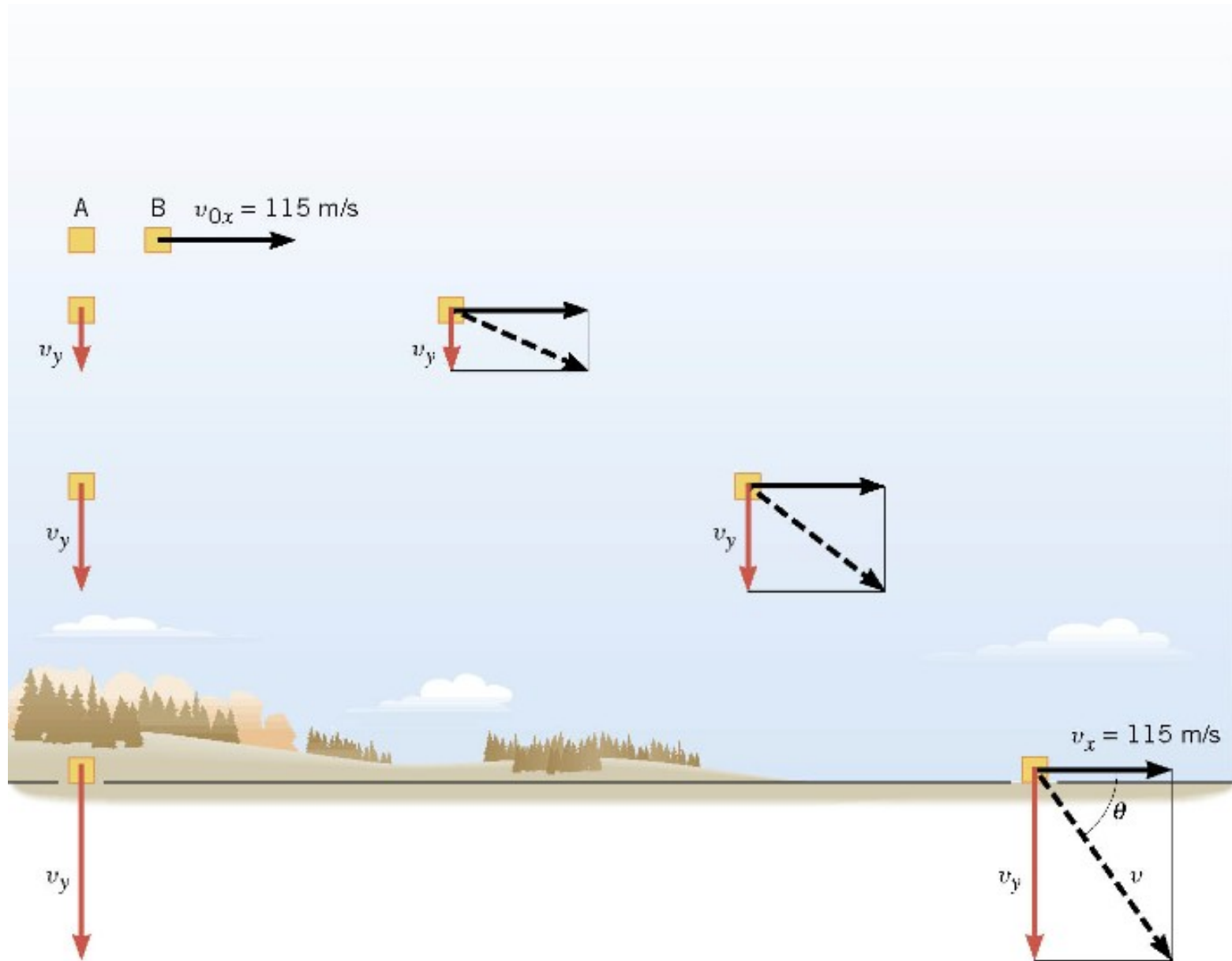
- Determine the time required for the care package to hit the ground. (use the y-axis free-fall equations)
- The final vertical velocity v_y (free-fall equation)
- The horizontal displacement (equation for motion at constant speed along X-axis)
- Optional: Using the components of the final velocity v_{0x} and v_y find the final velocity v (direction and magnitude).

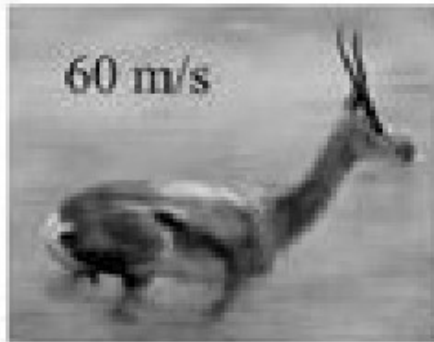


Will they hit at the same time ?

Will the final speed be the same?

Both packages are dropped at the same time.



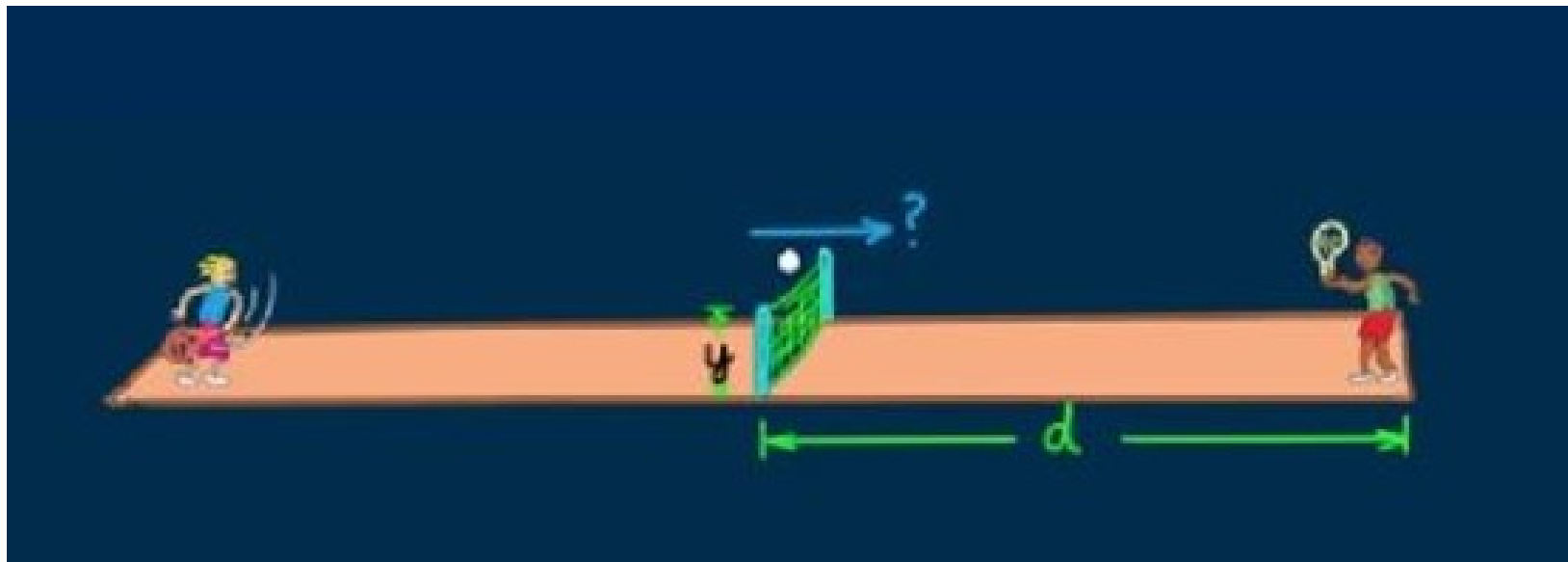


X	Y
$\vec{V}_x = 60 \text{ m/s}$	$\vec{V}_y = 0 \text{ m/s}$
$\vec{\Delta X} = ?$	$a = -9.8 \text{ m/s}^2$
	$\vec{\Delta y} = -30 \text{ m}$

30 m

$\vec{\Delta X} = ?$



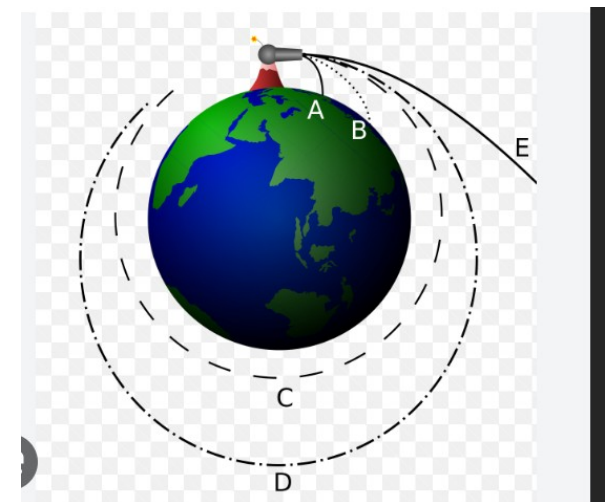


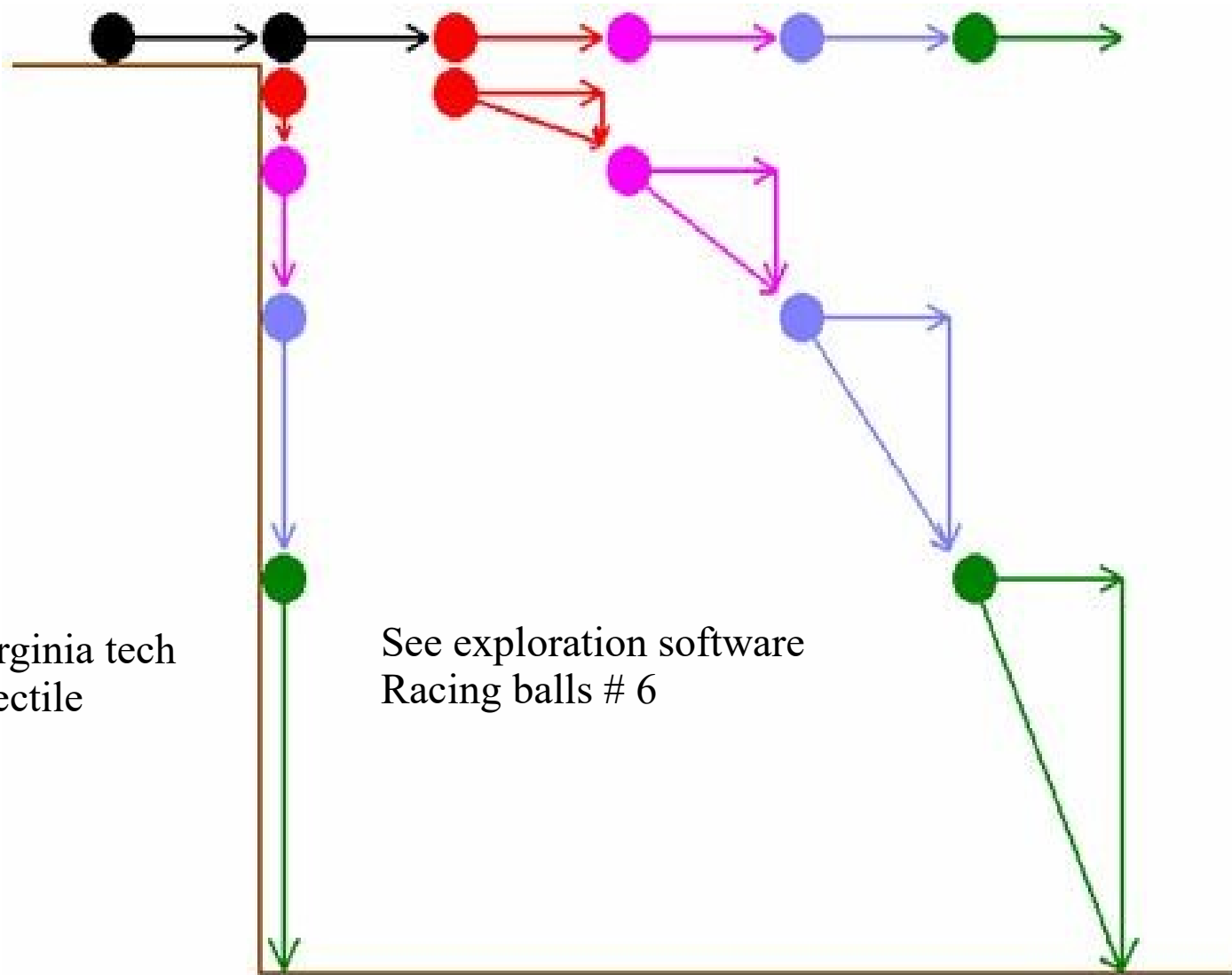
Source: Paul Hewitt (conceptual physics). $d = 10\text{m}$ and $y=1\text{m}$ what is the horizontal velocity to clear the net?

Source: Paul Hewitt (conceptual physics). The Earth is curving away : 8000m run and 5 m fall.

What is the minimum horizontal velocity for a launched from mountain so it does not crash (sic Newton)

1. An object falls 5 m in its first second of fall.
2. Earth's surface curves 5 m vertically for each 8000-m tangent.





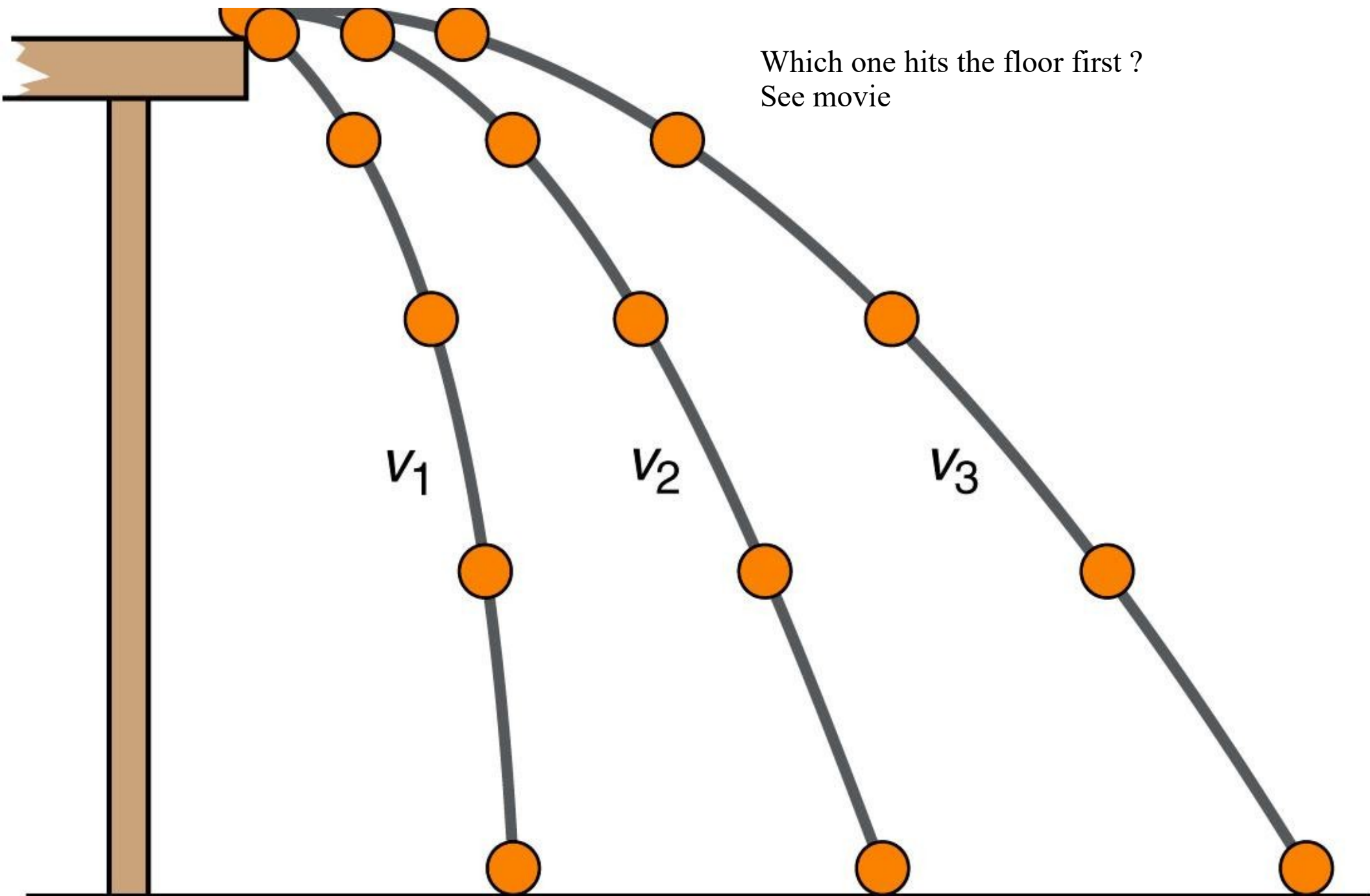
Video from virginia tech
free_fall_projectile

See exploration software
Racing balls # 6

https://www.walter-fendt.de/html5/phen/projectile_en.htm

1 motion but 2 independent components.

<https://www.youtube.com/watch?v=zMF4CD7i3hg>
shoot n drop



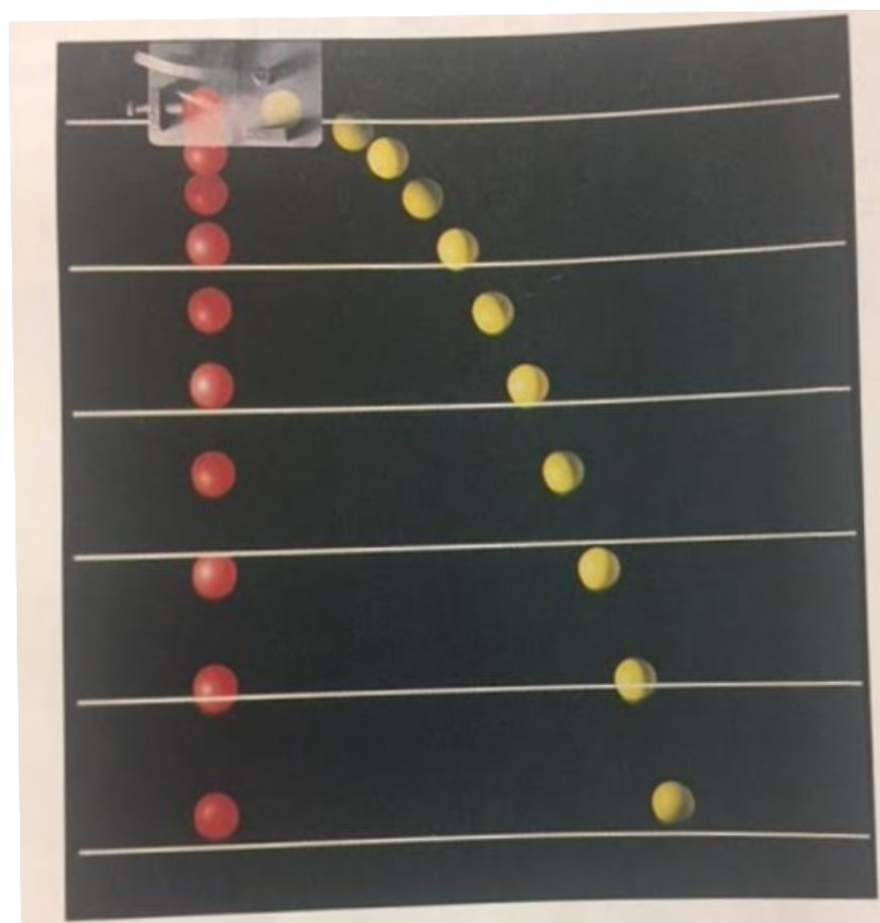
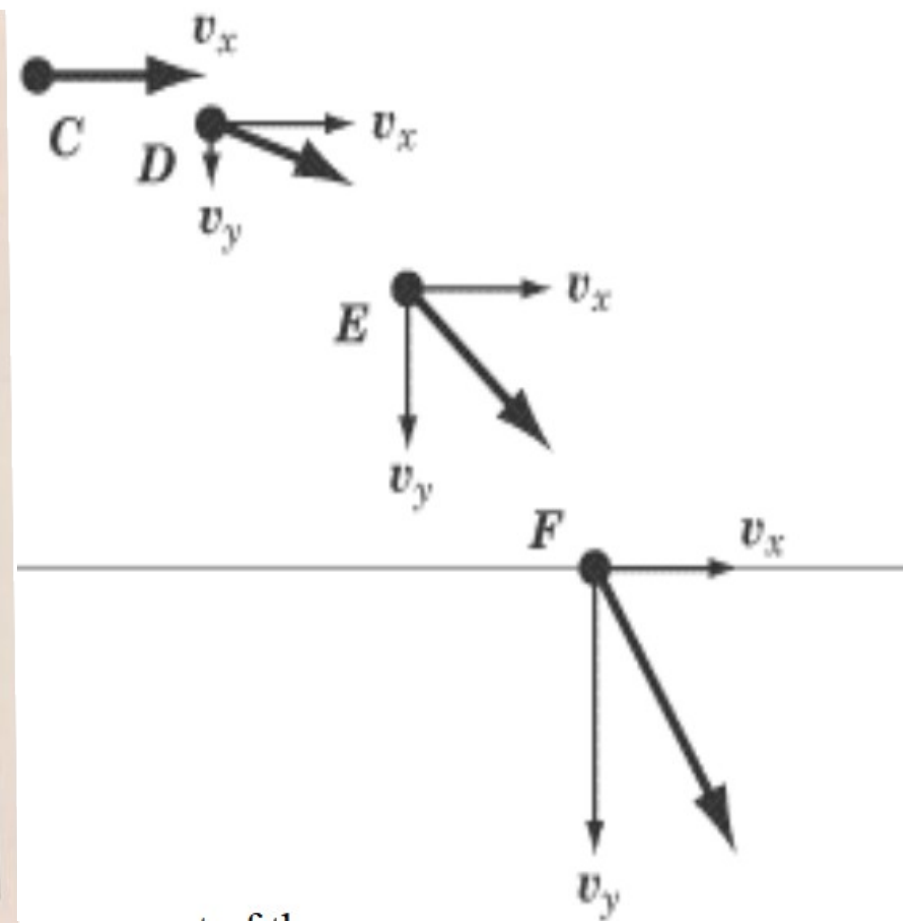


figure 3.16 A ball is dropped at the same time that a second ball is projected horizontally from the same height.



1) A bullet is fired horizontally with an initial velocity of $V_{ix}=200\text{m/s}$ (V_x) at a target $\Delta x=2500\text{m}$ From the rifle.

A) How much time is required for the bullet to reach the target?

Hint: Use the x-axis equations. along the horizontal, there is no acceleration, the velocity stays the same)

B) Using the approximation $g= -10\text{m/s/s}$, how far does the bullet fall in time. (so Δy)

Hint: Use the y-axis equations. along the vertical, acceleration is g and the initial velocity is 0

C) Find the final y-component V_y

D) me: the final velocity V has components V_x and V_y . Draw the vector in the X-Y coordinate system.

Find its magnitude and its direction. $V = \underline{\hspace{1cm}} \text{ m/s @ } \underline{\hspace{1cm}}$

2) A ball rolls off a shelf with a horizontal velocity of $V_{ix}=6\text{m/s}$. At what horizontal distance x From the shelf does the ball land if it takes 4s to reach the ground ?

Hint: Use the x-axis equations

3) A ball rolls off a table with a horizontal velocity of $V_{ix}=4\text{m/s}$ / If it takes 0.4s for the ball to reach The ground , how high above the floor is the tabletop ?

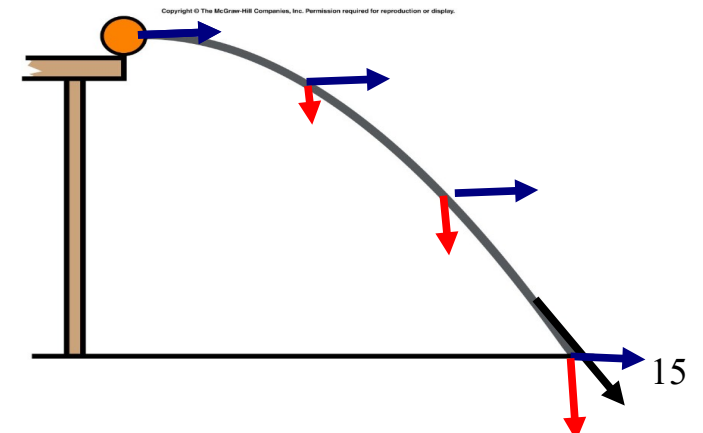
Hint: $V_{iy}=0$ Use the y-axis equations. V_{ox} is extra information.

4) A boll rolls off a table with a horizontal velocity of $V_{ix}=5\text{m/s}$. It it takes 0.4s for it to reach the floor:

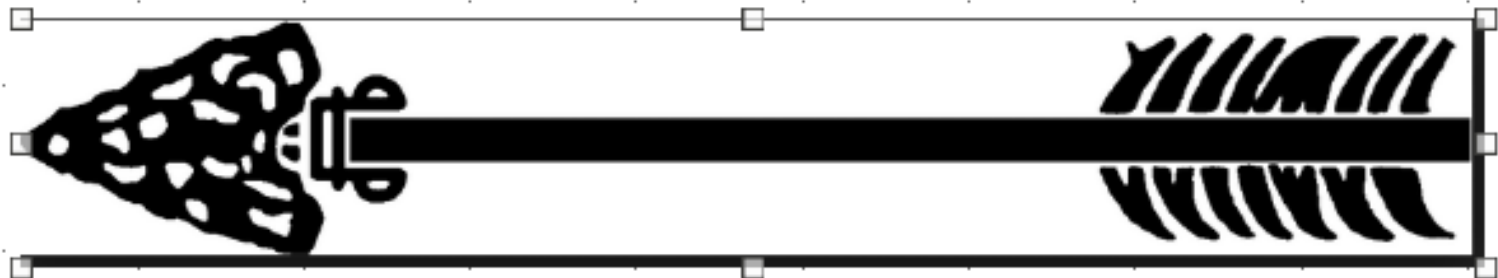
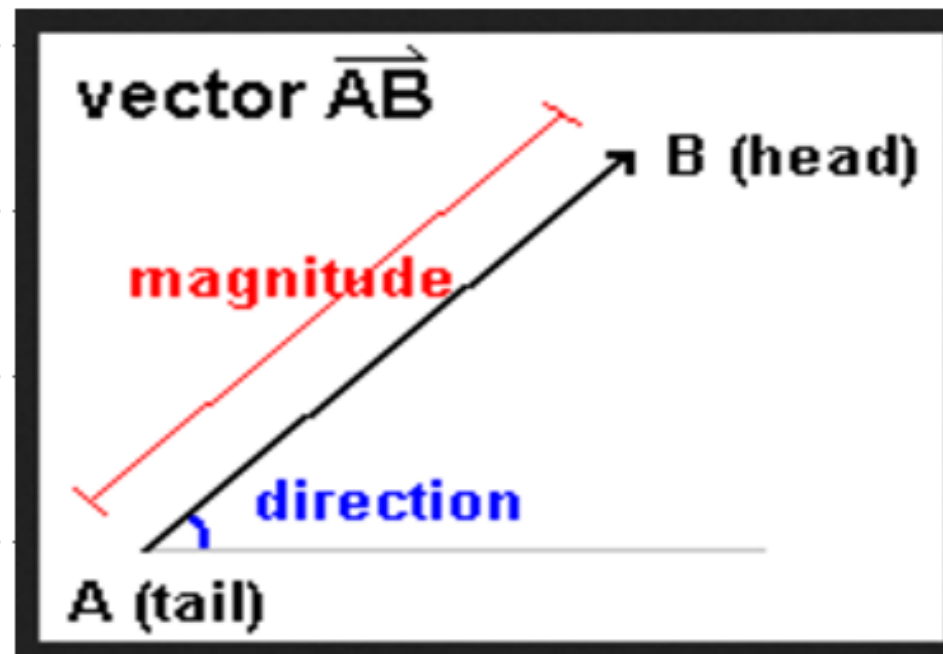
A) What is the vertical component of the ball's velocity V_y just before it hits the floor ($g=-10$). Use y-axis equations

B) What is the horizontal component of the ball's velocity just before it hits the floor ? V_x

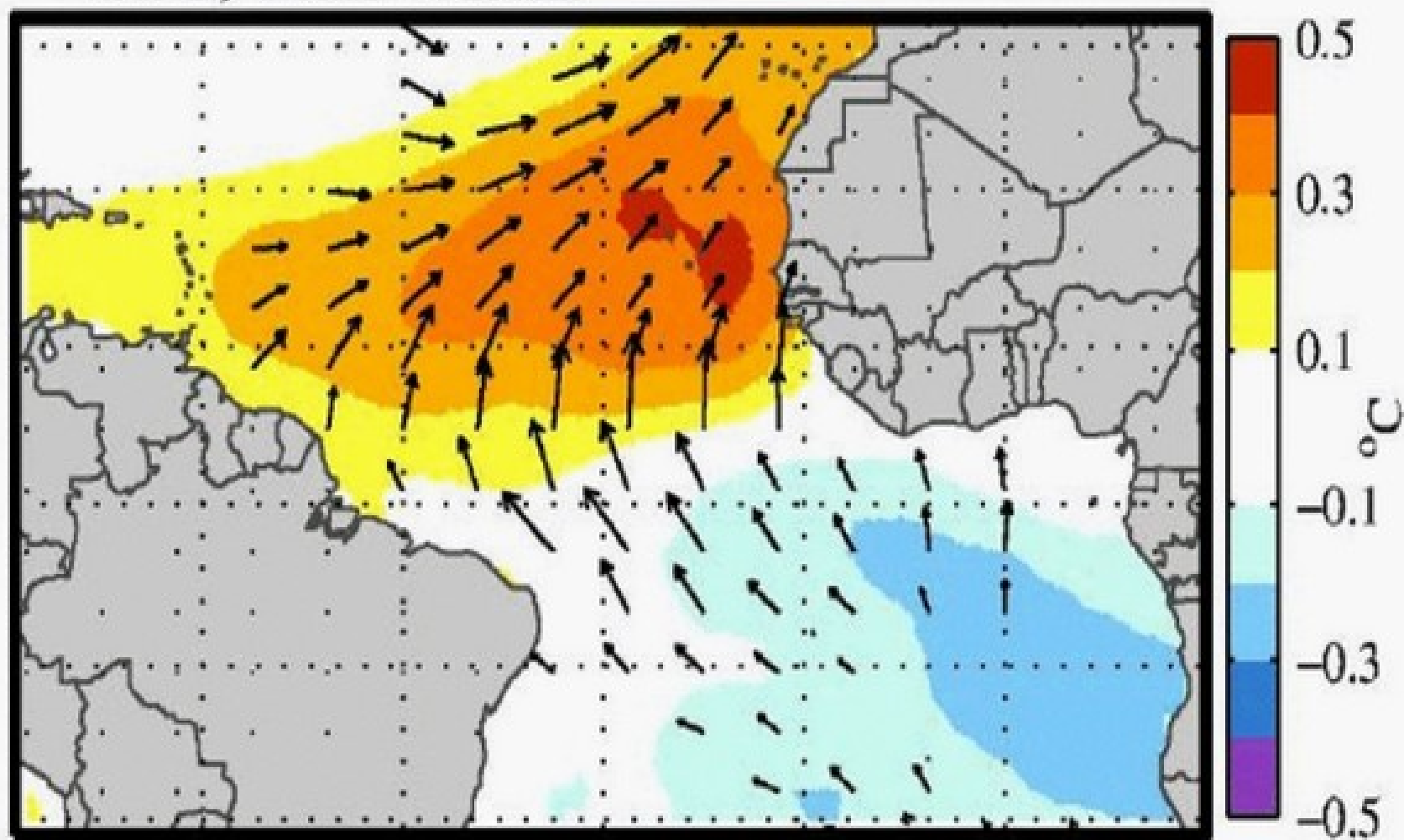
Use the x-axis equations C) Find V_x and V_y at $t=0.4\text{s}$. Find the final velocity V (components V_x and V_y)



VECTORS



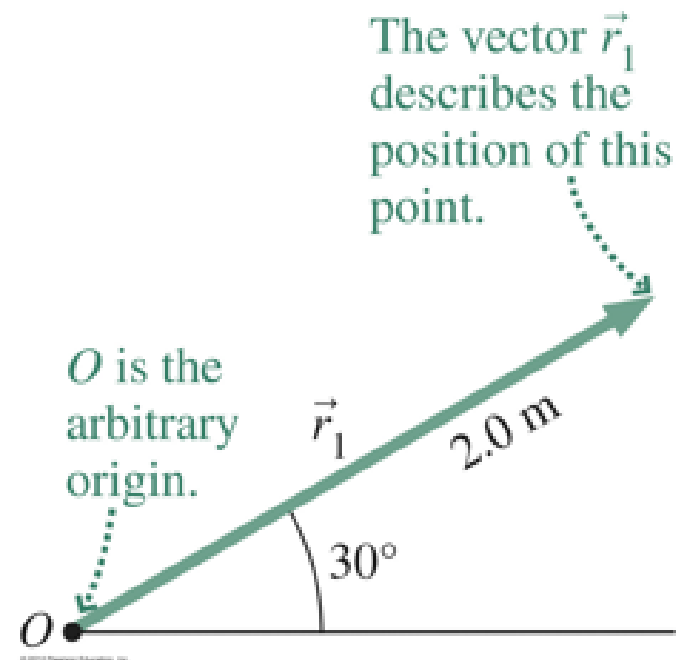
SST, 10m Winds



Precipitation

Scale: 0.5 m/s = $\longrightarrow$

- A **vector** is a quantity that has both magnitude and direction.
 - In two dimensions it takes two numbers to specify a vector.
 - In three dimensions it takes three numbers.
 - A vector can be represented by an arrow whose length corresponds to the vector's magnitude.
- Position is a vector quantity.
 - An object's position is specified by giving its distance from an origin and its direction relative to an axis.
 - Here $\vec{r}_1$ describes a point 2.0 m from the origin at a 30° angle to the axis.

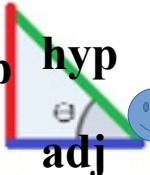


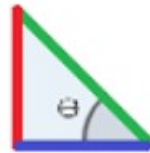
Review:

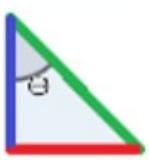
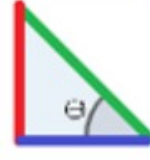
<https://www.khanacademy.org/math/geometry/hs-geo-trig/hs-geo-trig-ratios-intro/v/basic-trigonometry>

SOH CAH TOA

adj 

SOH   

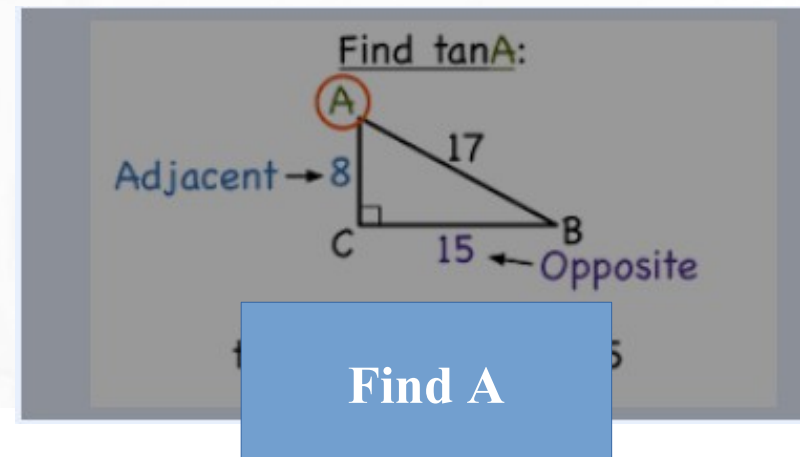
CAH  

TOA  

$\sin \theta = \frac{\text{Opposite}}{\text{Hypotenuse}}$

$\cos \theta = \frac{\text{Adjacent}}{\text{Hypotenuse}}$

$\tan \theta = \frac{\text{Opposite}}{\text{Adjacent}}$



(0) Sum angles = 180°

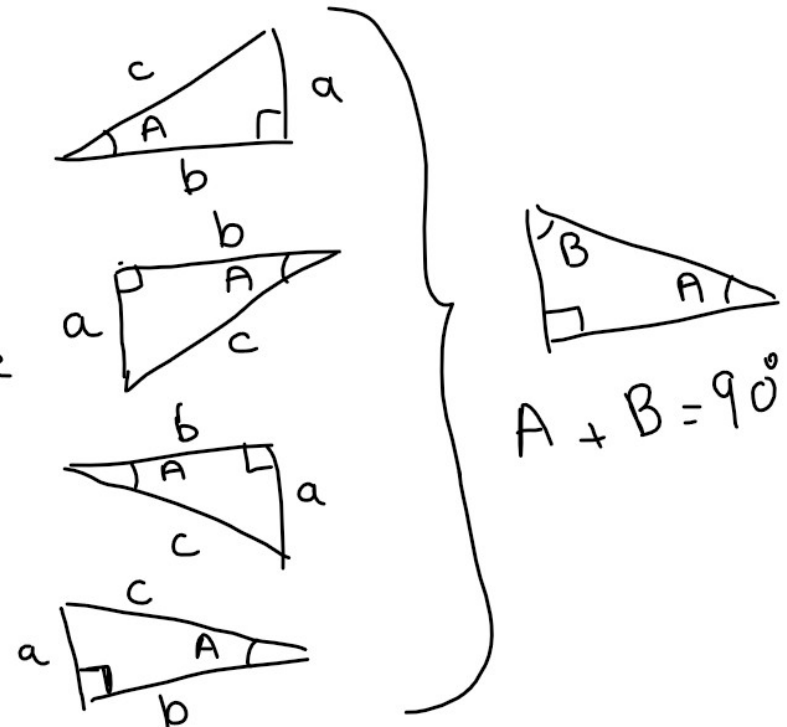
(1) $a^2 + b^2 = c^2$

(2) $A = \tan^{-1}\left(\frac{a}{b}\right)$

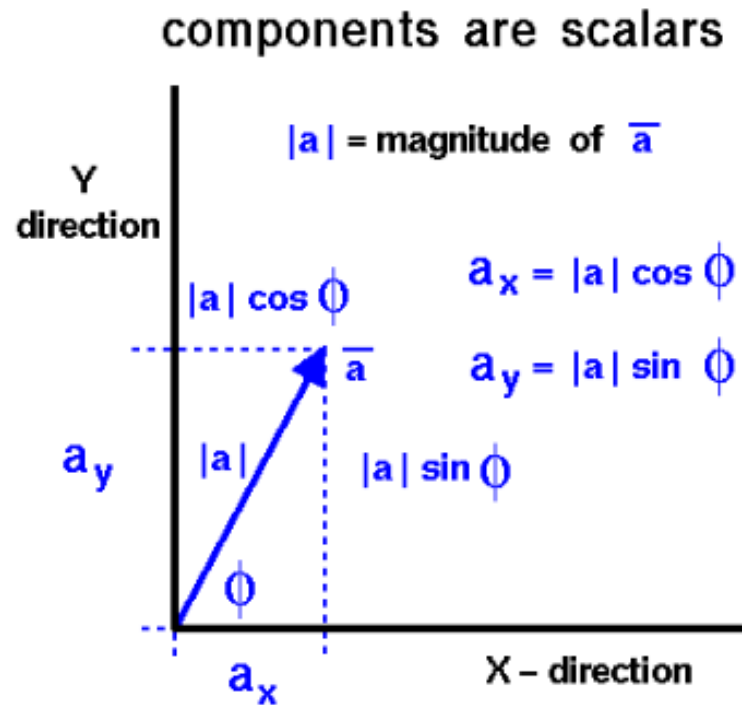
(3) $\frac{a}{b} = \frac{\tan(A)}{1}$ } slope

(4) $\frac{b}{c} = \frac{\cos(A)}{1}$

(5) $\frac{a}{c} = \frac{\sin(A)}{1}$



A **vector quantity** has both **magnitude** and **direction**.



See the tutorials in my dropbox (vector part 1 and trigonometry)

If the vector is in the first quadrant, just use cosine for the x-component and sine for the y-component

https://phet.colorado.edu/sims/html/vector-addition/latest/vector-addition_all.html

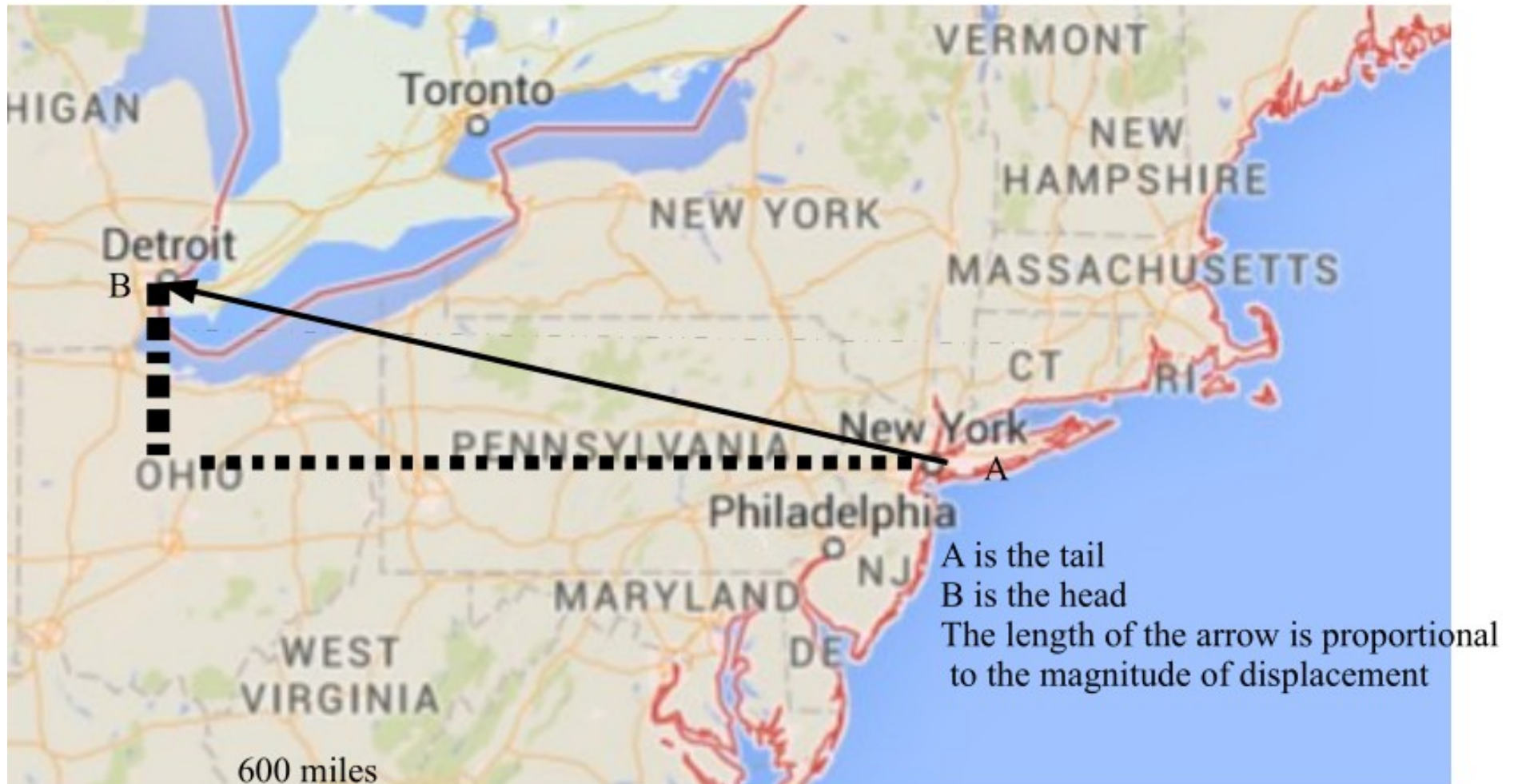
<https://ophysics.com/k3.html>

Vector displacement AB

Vectors come in parts. A vector has 2 components in 2D.

A horizontal component along the x-axis or W-E direction
and a vertical component along the y-axis or the S-N direction

Here you can find how far in the West direction and how far in the North direction



Find the components of the vector displacement.

The magnitude is 600 miles and the angle is 10 degree. Use trig.

Displacement AB= 600 miles @ 10 degrees N of W

2 notations:

#1 standard notation (polar graph)

AB=600 miles @ 170 degrees

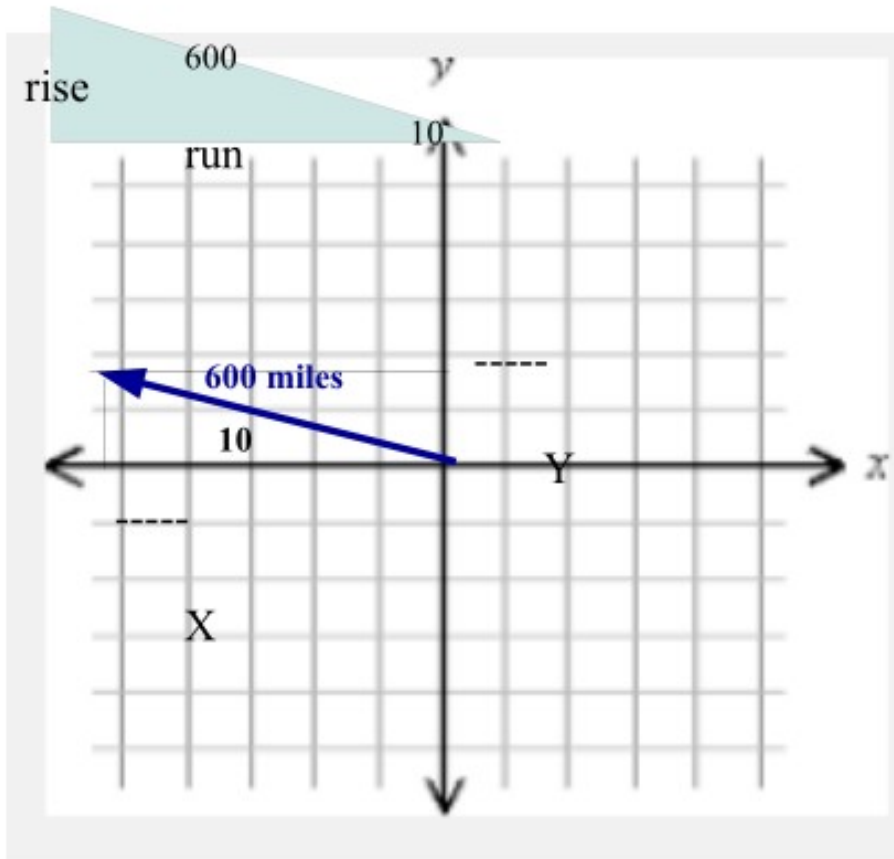
#2. Cartesian coordinates

The angle is 170 degrees

$X = 600 \cos(170) = -591$ miles

$Y = 600 \sin(170) = 104$ miles

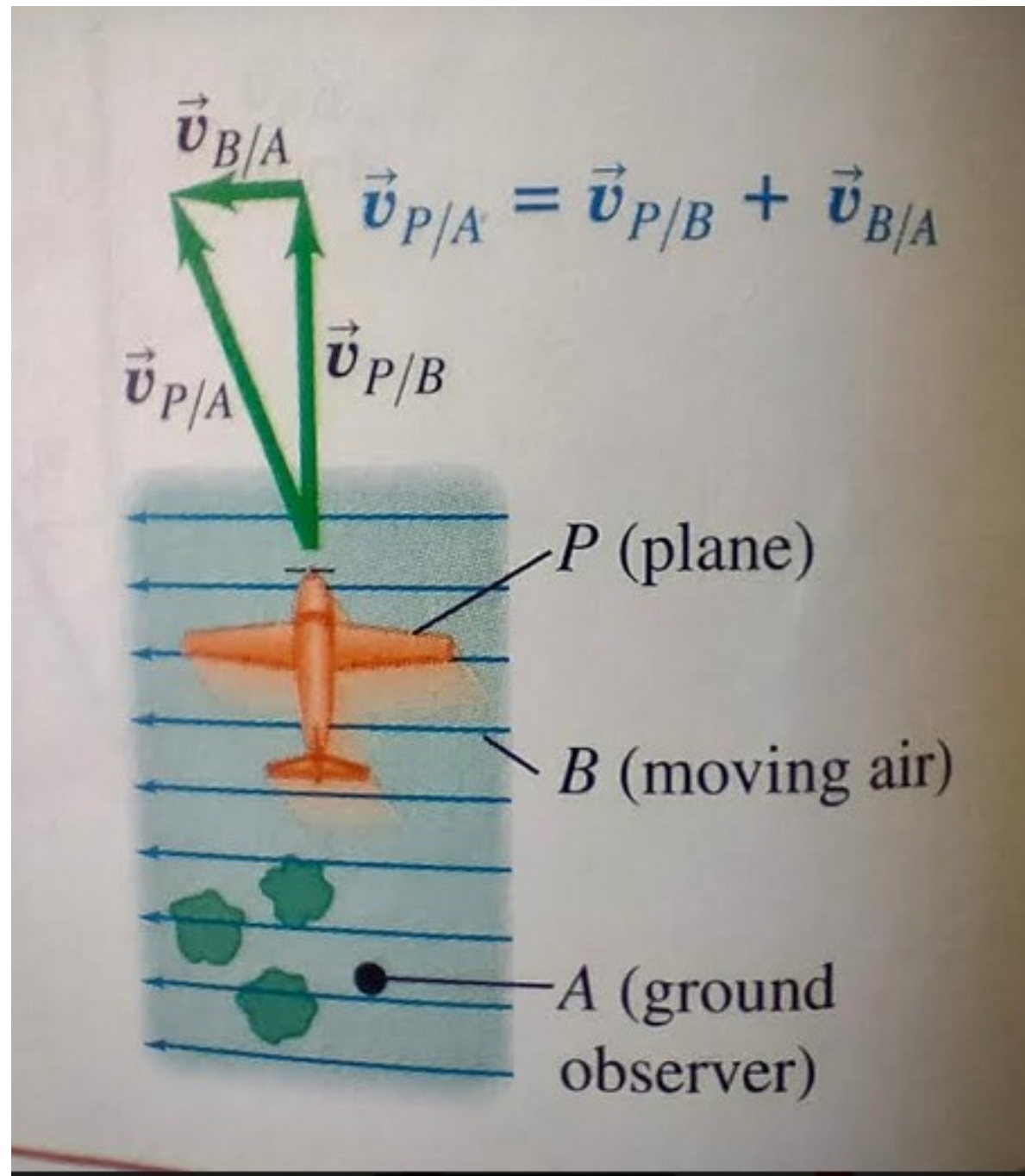
AB(-591,104)



So the components of the displacement are $X = -591$ miles $Y = 104$ miles
The object moved 591 @ west and 104 miles at north.

See vectors website

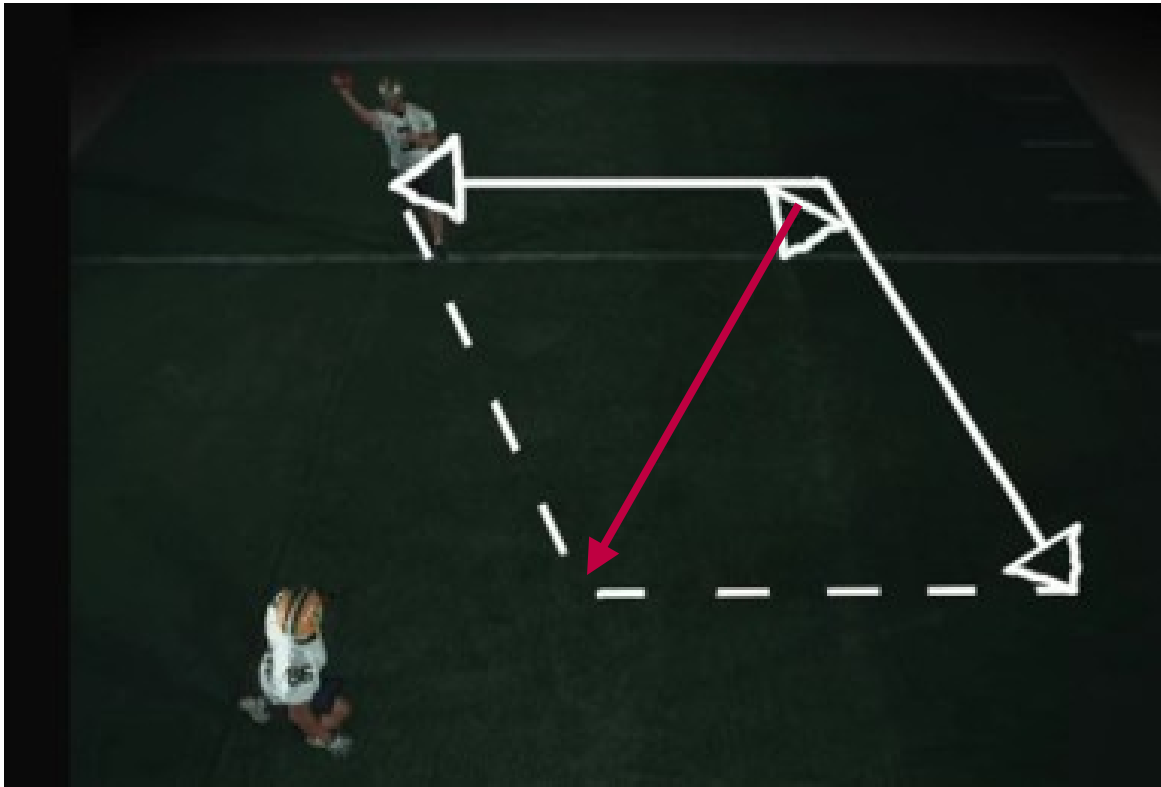
Note the tail to head method. Start from tail $\rightarrow$ head



Relative to Gnd
The velocity of
The plane has
2 components.

North and west.

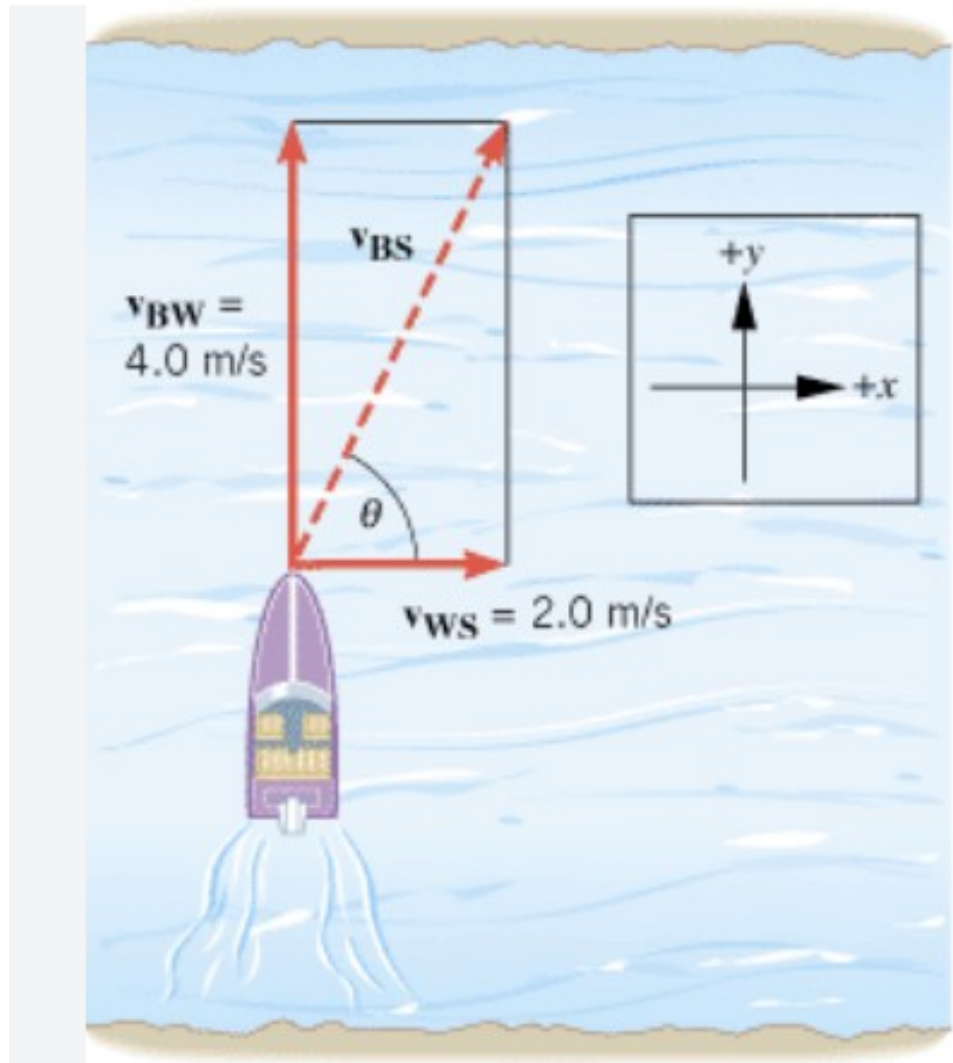
The motion has
2 components
Independent from
Each other

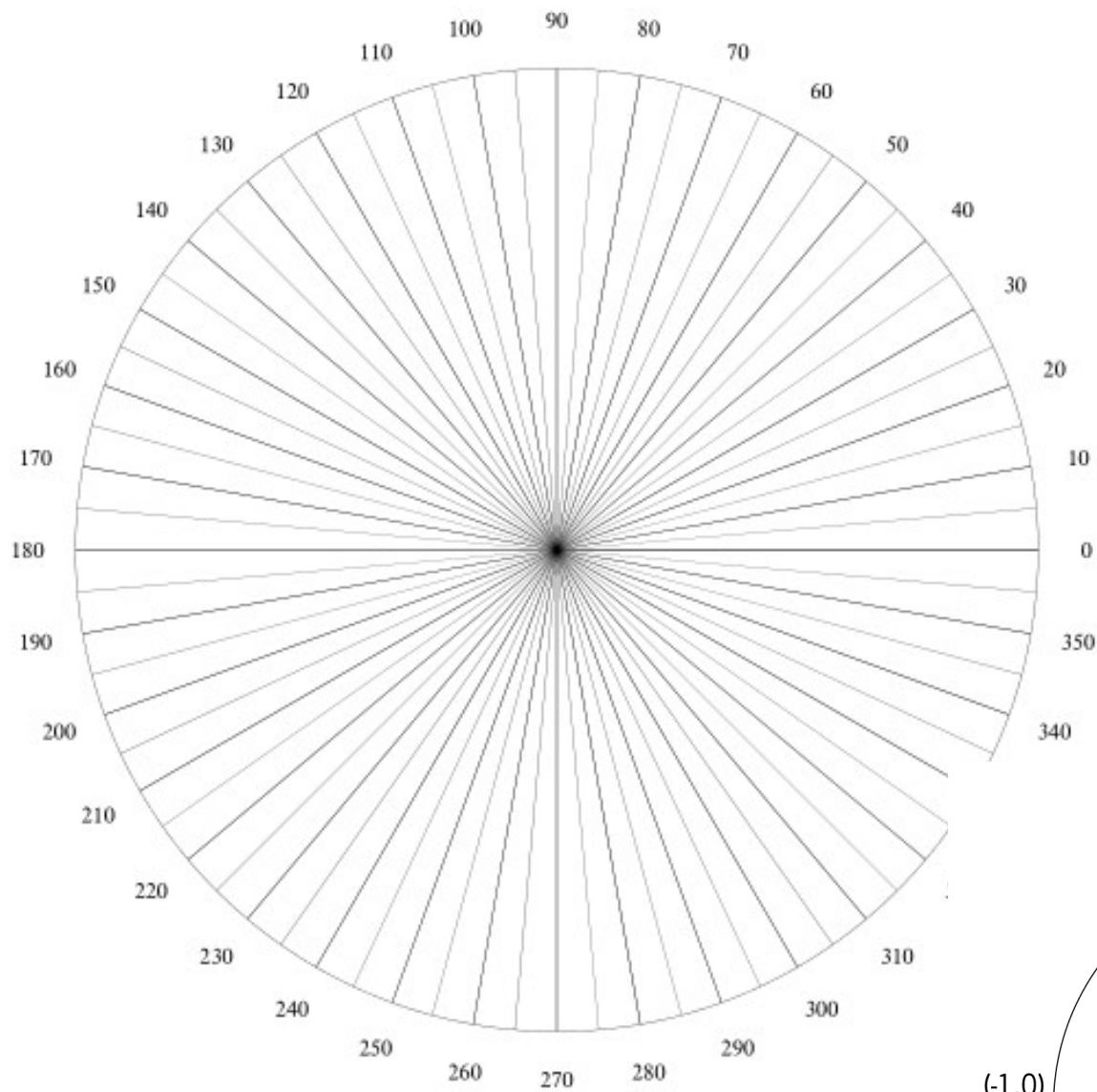


The vector velocity of the ball has 2 components (perpendicular)
The motion of the ball is defined by the final velocity
= the vector sum of the 2 components

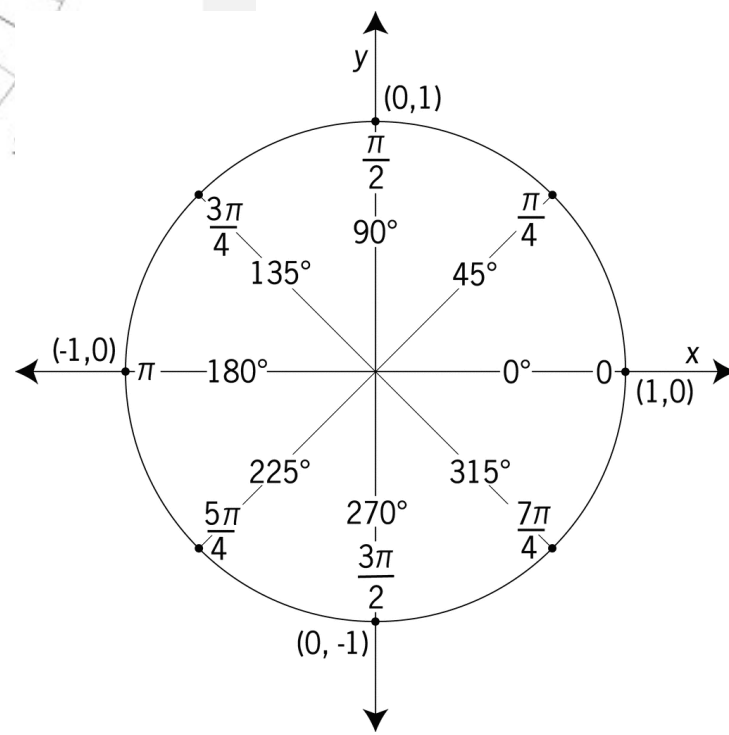
If the river is 1,000 wide. How long to reach the other side

<https://ophysics.com/k11.html>





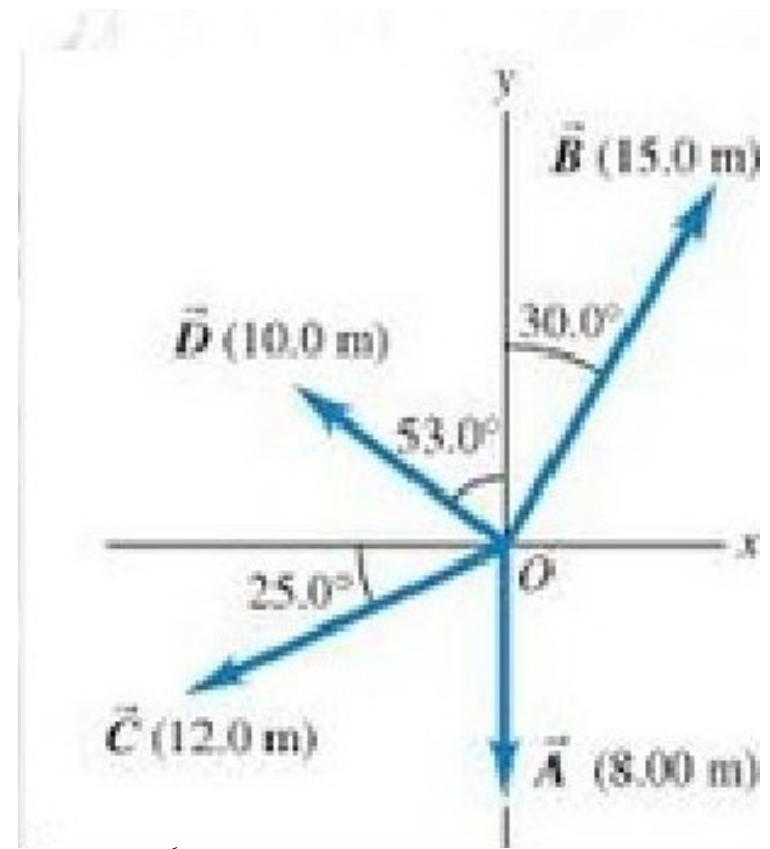
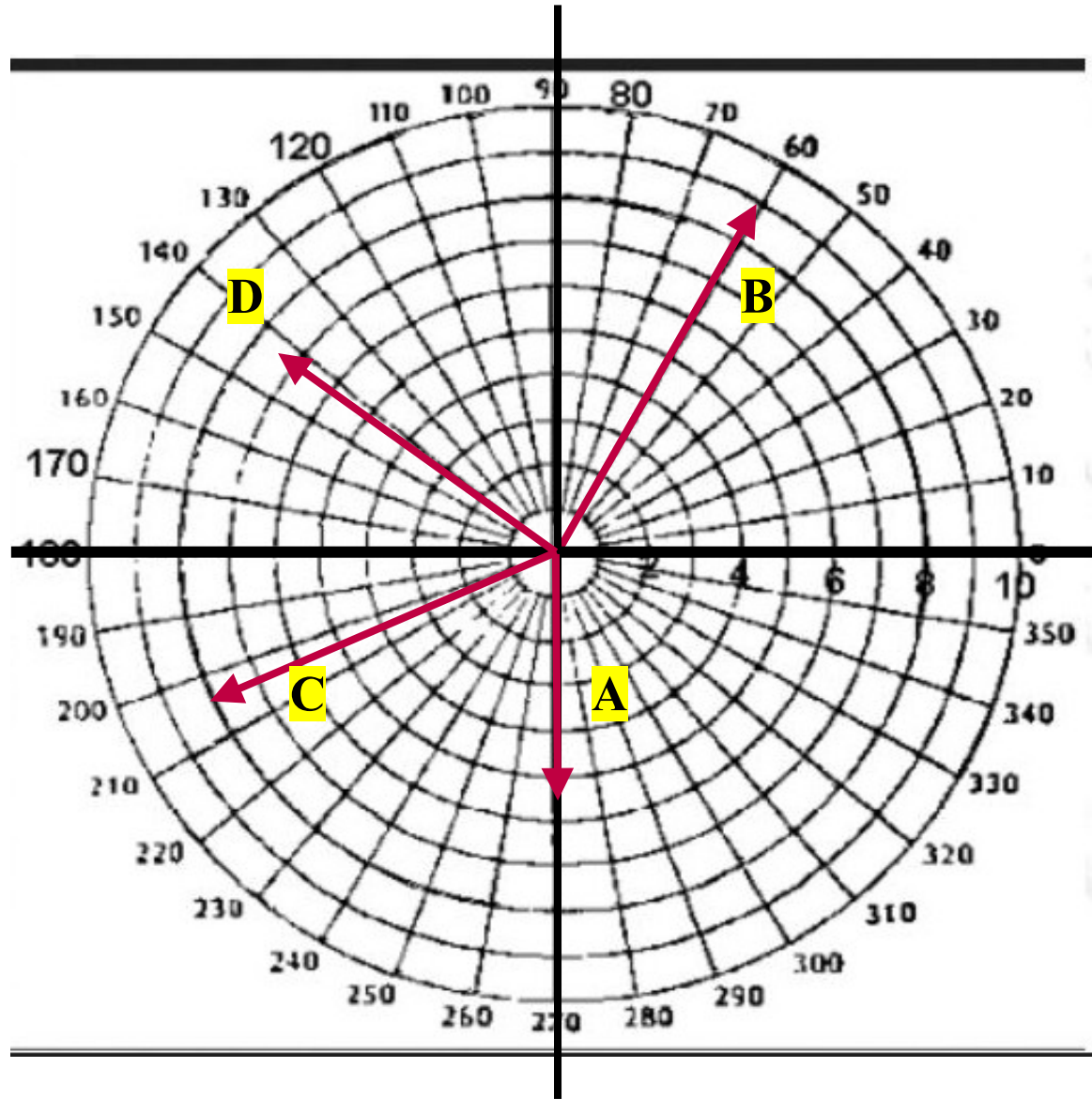
What if the vector is not
In the first quadrant?



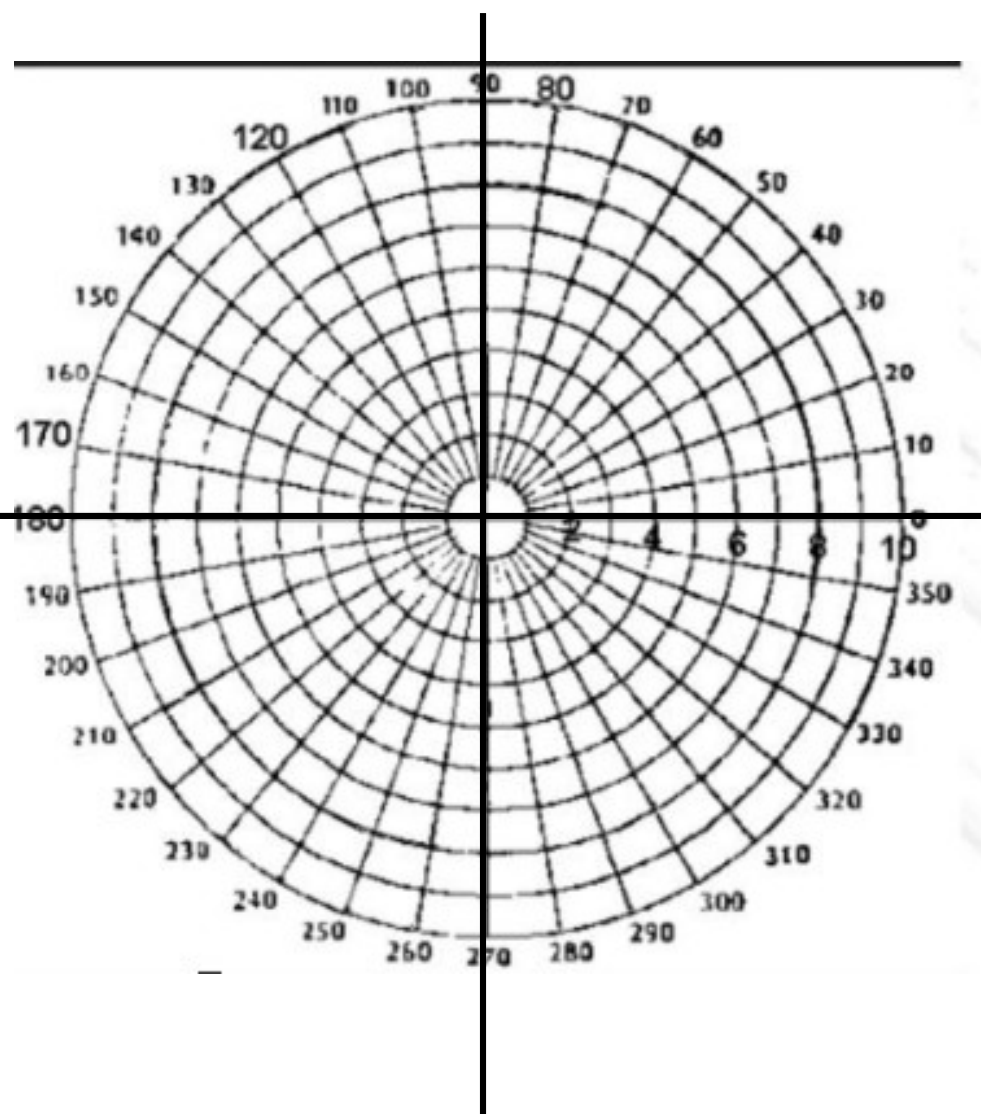
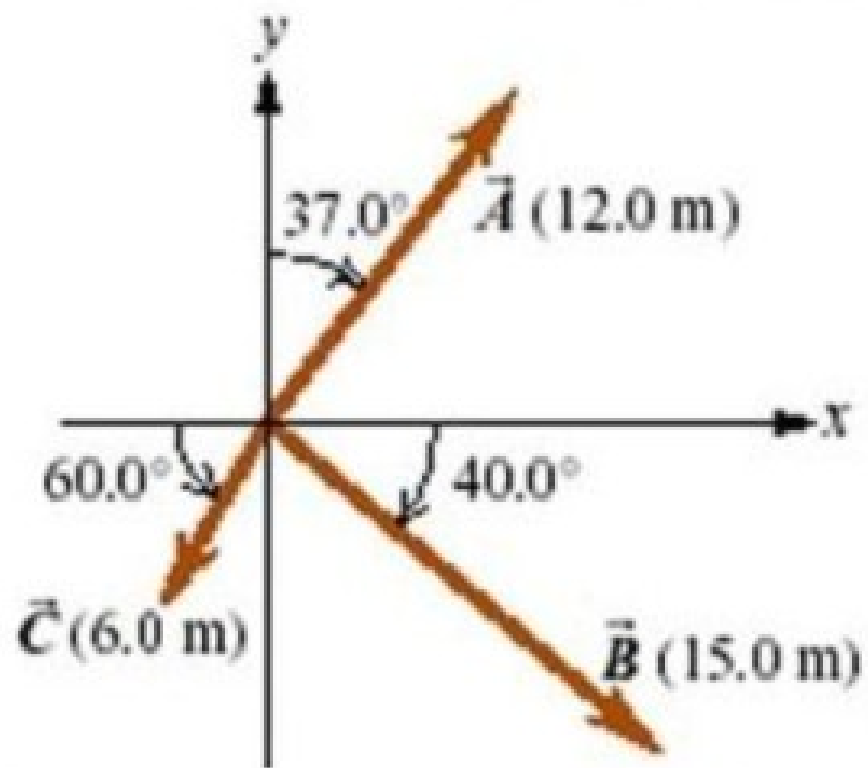
Finding the components:

→ How much of the vector is along X

→ How much of the vector is along Y



If you cut and paste
The vector in the polar graph,
X-component is cosine
Y-component is the sine.



) A velocity vector has components 36 m/s westward and 22 m/s northward. What are the magnitude and direction of this vector?

9) A boy jumps with a velocity of magnitude 20.0 m/s at an angle of 25.0° above the horizontal. What is the horizontal component of the boy's velocity?

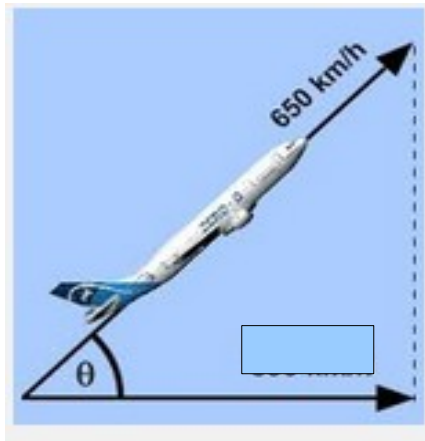
A) 15.6 m/s

B) 12.6 m/s

C) 18.1 m/s

D) 8.45 m/s

E) 9.33 m/s



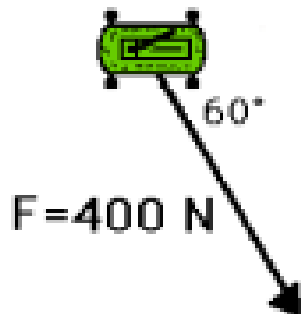
How fast the plane is going up ? (vertical component of the velocity)

How fast is plane is moving forward?

(how long it will take for the plane to be over your head)



What are the components of the force?

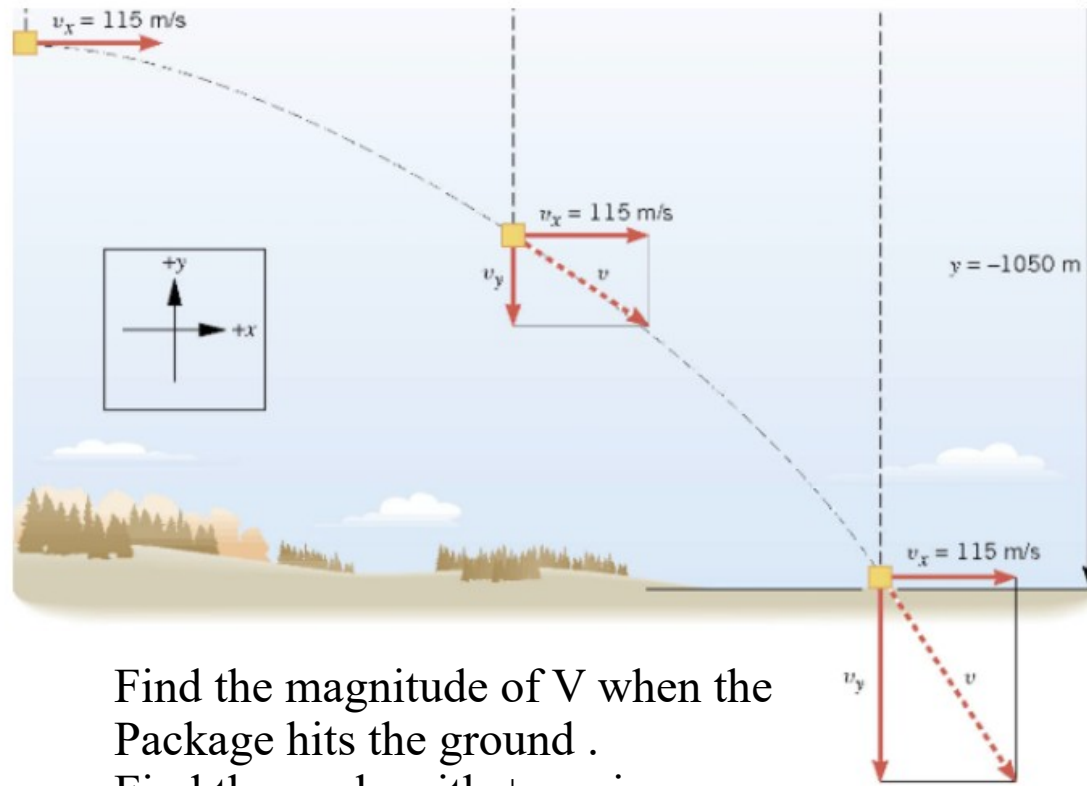


How hard the wind is pushing forward?

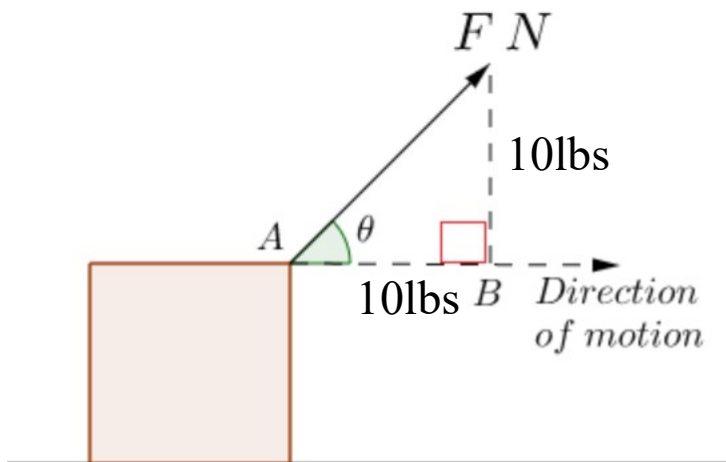
How hard the wind is pushing side way?

Example: UsainBolt. Starter block

How to find an angle ? Magnitude? Using the components.



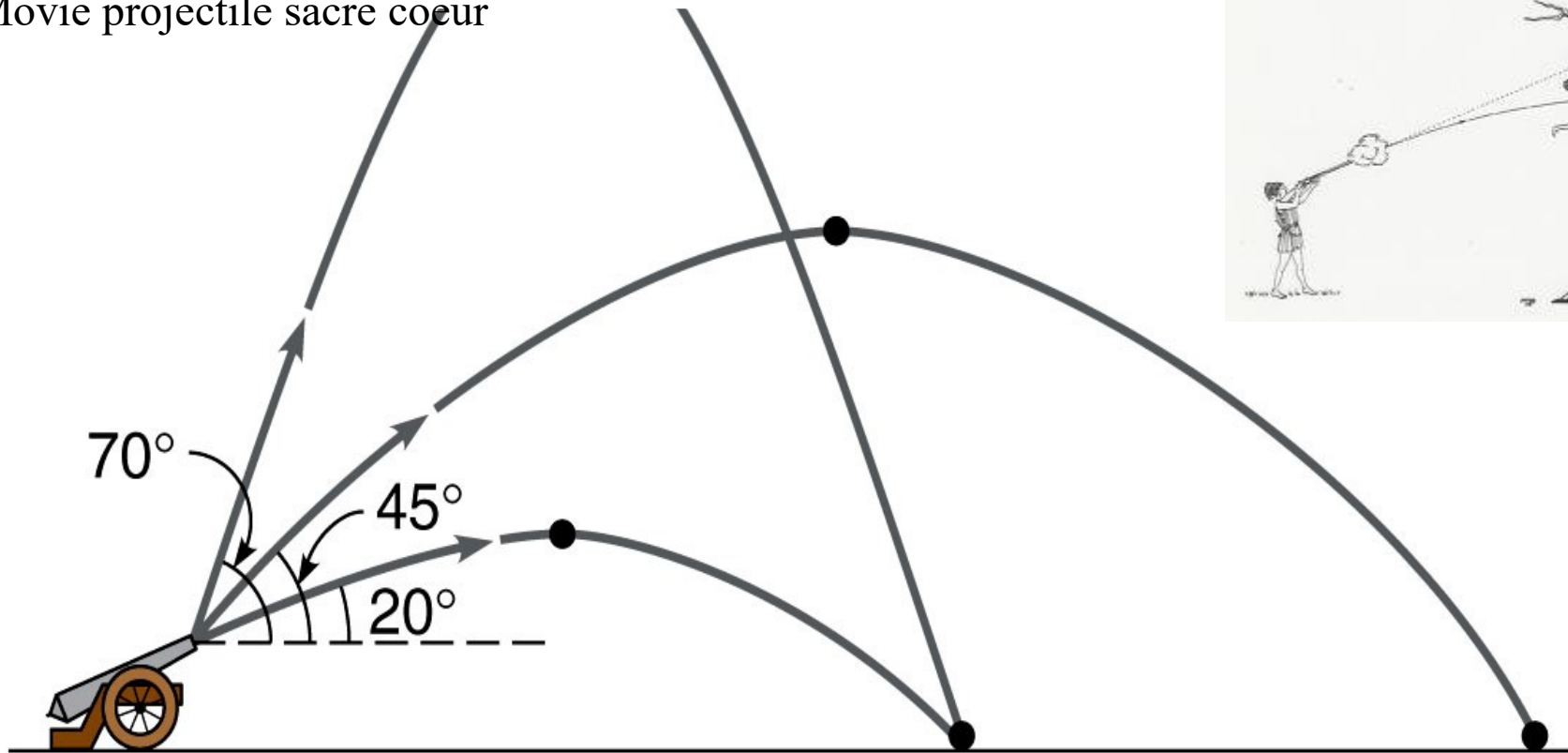
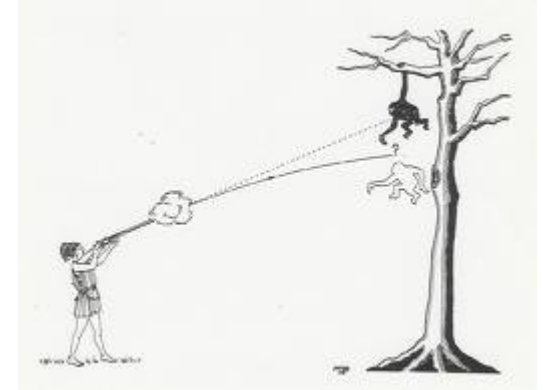
Find the magnitude of V when the Package hits the ground .
Find the angle with $+x$ -axis



Find angle and magnitude

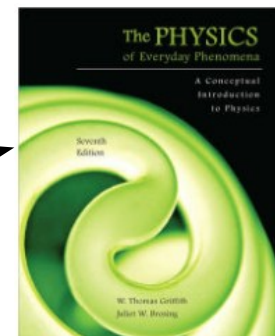
Projectile motion

Movie projectile sacre coeur



References for this unit:

- The physics of everyday phenomena: a conceptual introduction to physics by W. Thomas Griffith, McGraw-Hill
- Physics / Edition 8 by John D. Cutnell, Kenneth W. Johnson, Cutnell



Remember that acceleration is force per unit mass. As long as there is a force acting -. acceleration
The acceleration due to gravity is the weight per unit mass.

A ball is launched with an initial velocity $\vec{v}_0$ as shown. Which one of the following arrows best represents the direction of the acceleration at point A? B? C?

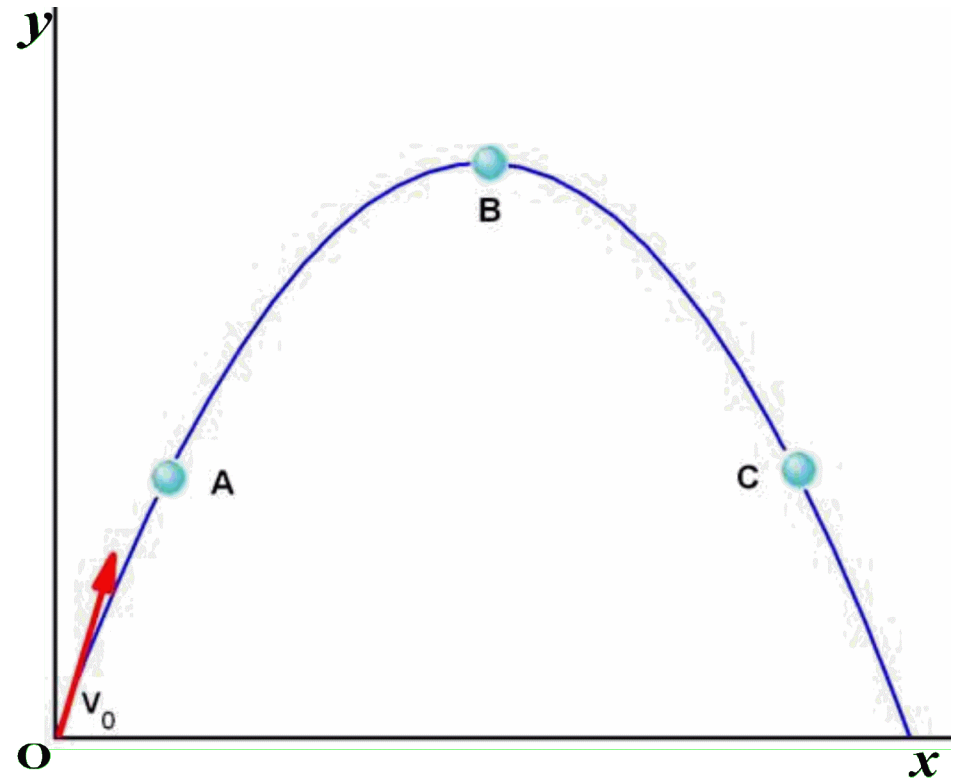
a) 

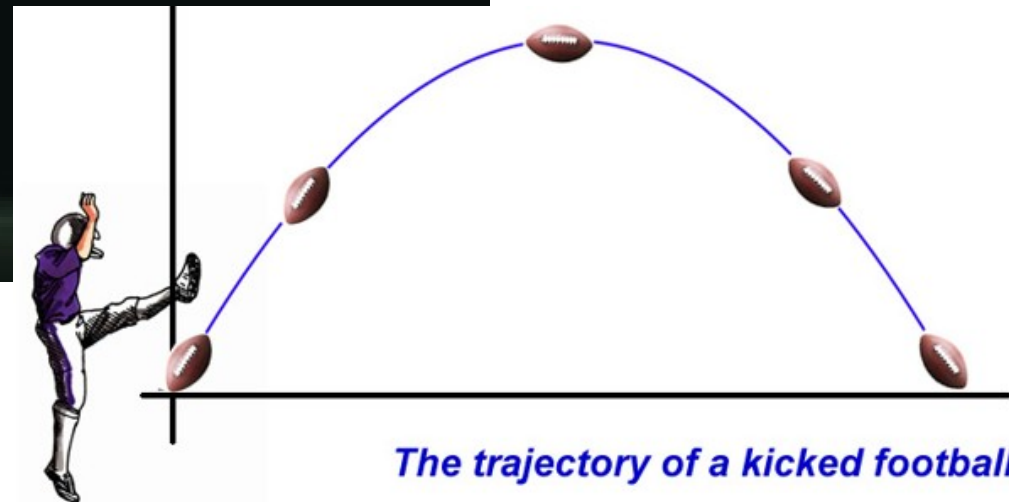
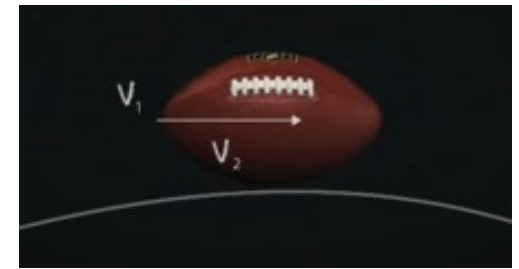
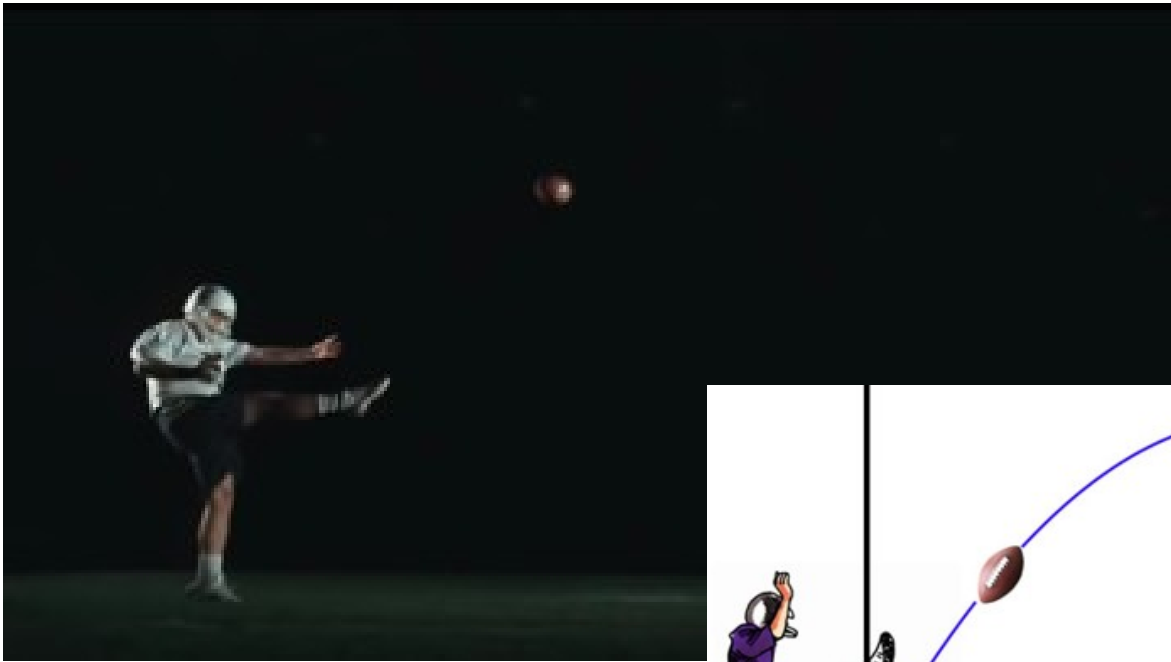
b) 

c) 

d) 

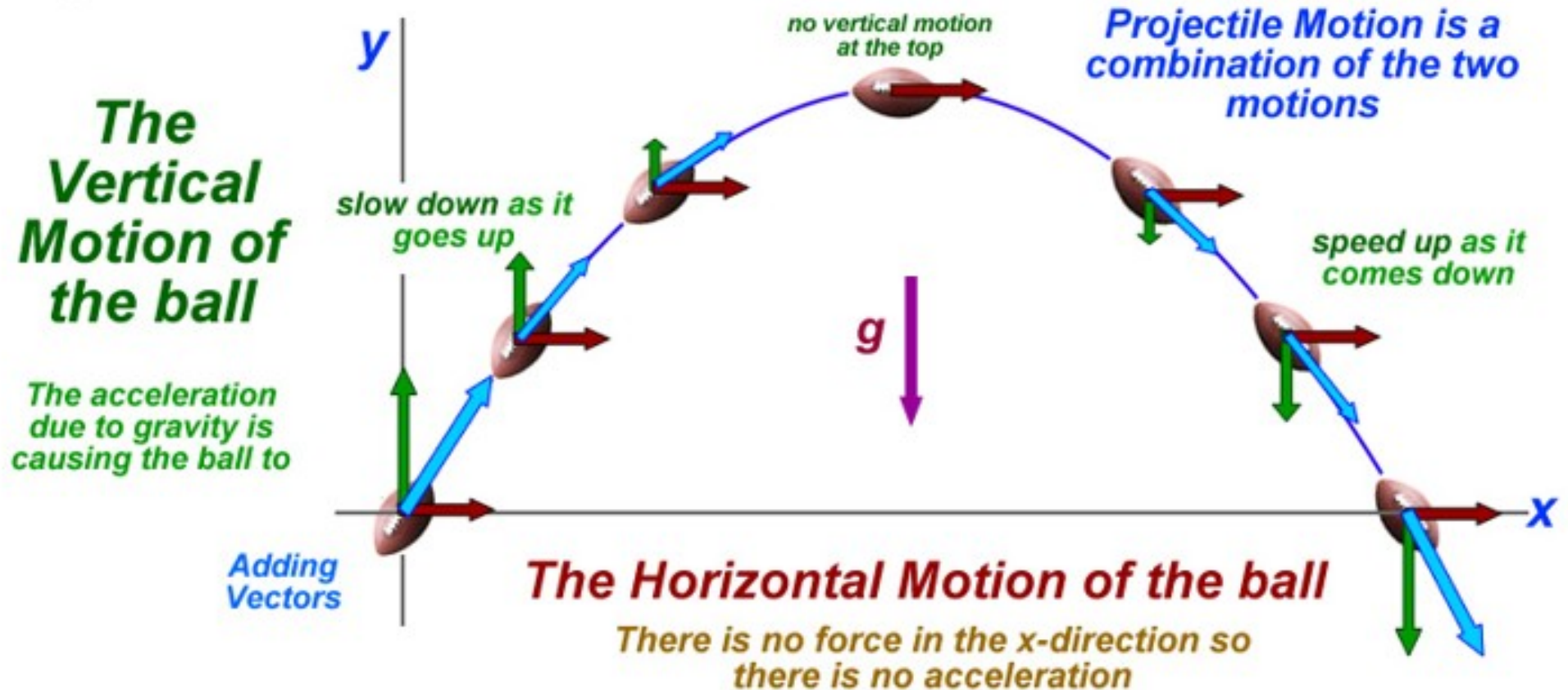
e) The acceleration at point A is zero m/s^2 .





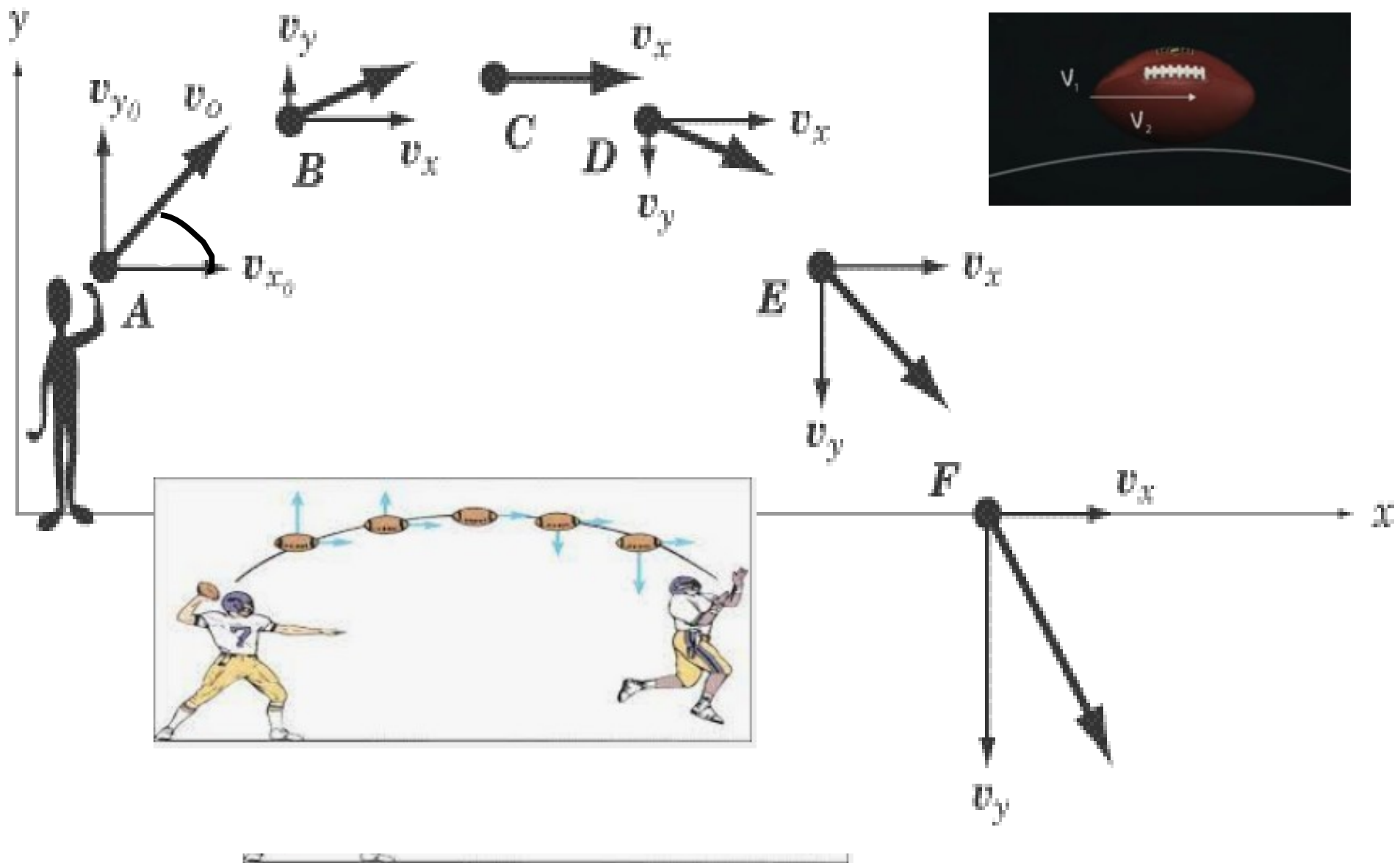
The initial velocity has 2 components
 Forward (horizontal) and vertical.
 The vertical component decreases to 0
 At apogee and the horizontal component
 Stays constant.

Projectile Motion - A Vector Perspective



Source:

<http://sdsu-physics.org/physics180/physics180A/units/unit1/chapter3.html>

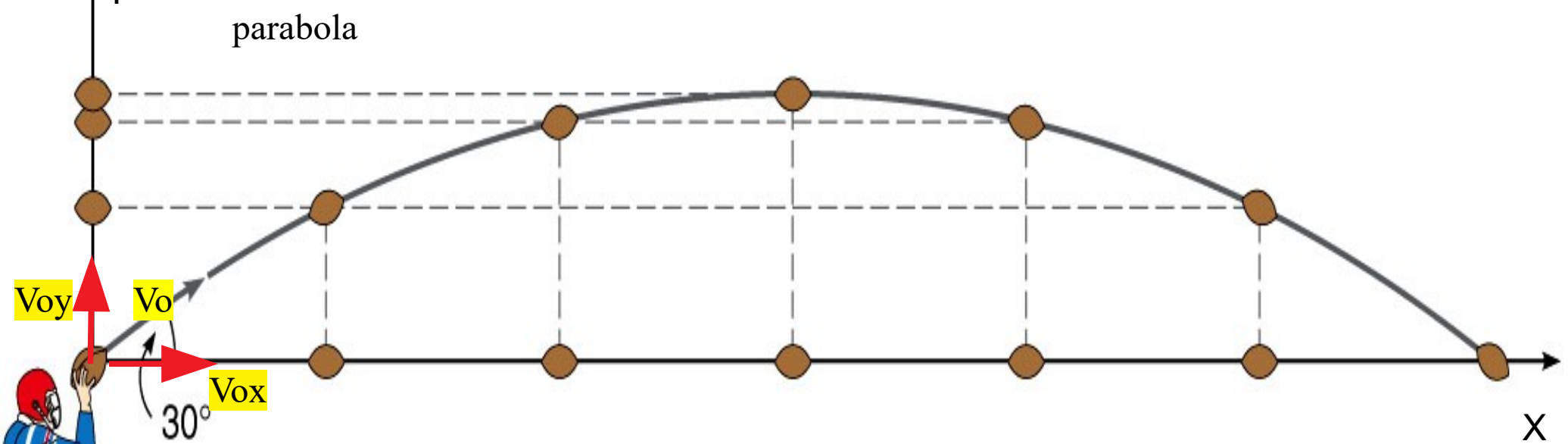


Along the horizontal (x-axis) the x- component of the Velocity stays the same. $v_{ix} = v_x$

Along the vertical (y-axis) the y-component of the velocity Changes at a constant rate : $v_y = v_{iy} - 10t$ so decreases at a rate of 10m/s/s

$$v_{ix} = v_0 \cos(\theta)$$

$$v_{iy} = v_0 \sin(\theta)$$



Along vertical (Y-axis)

Free-fall UP and DOWN

Acceleration is -10m/s^2

Δy is the vertical displacement (y-axis)

We suppose initial position is 0.

V_{iy} is the initial velocity along y

(V_{iy} is the y-component of V_i the initial velocity)

$$V_{oy} = V_o \sin(30)$$

$$\Delta Y = V_{oy} t - 5t^2$$

$$V_y = V_{oy} - 10 t$$

$$V_y^2 = V_{oy}^2 - 20\Delta Y$$

Along horizontal no acceleration

$a = 0$

Δx is the horizontal displacement (x-axis)

We suppose initial position is 0.

V_{ix} is the initial velocity along x

(V_{ix} is the x-component of V_i the initial velocity)

$$V_{ox} = V_o \cos(30)$$

$$\Delta x = V_{ox} t$$

$$V_x = V_{ox}$$

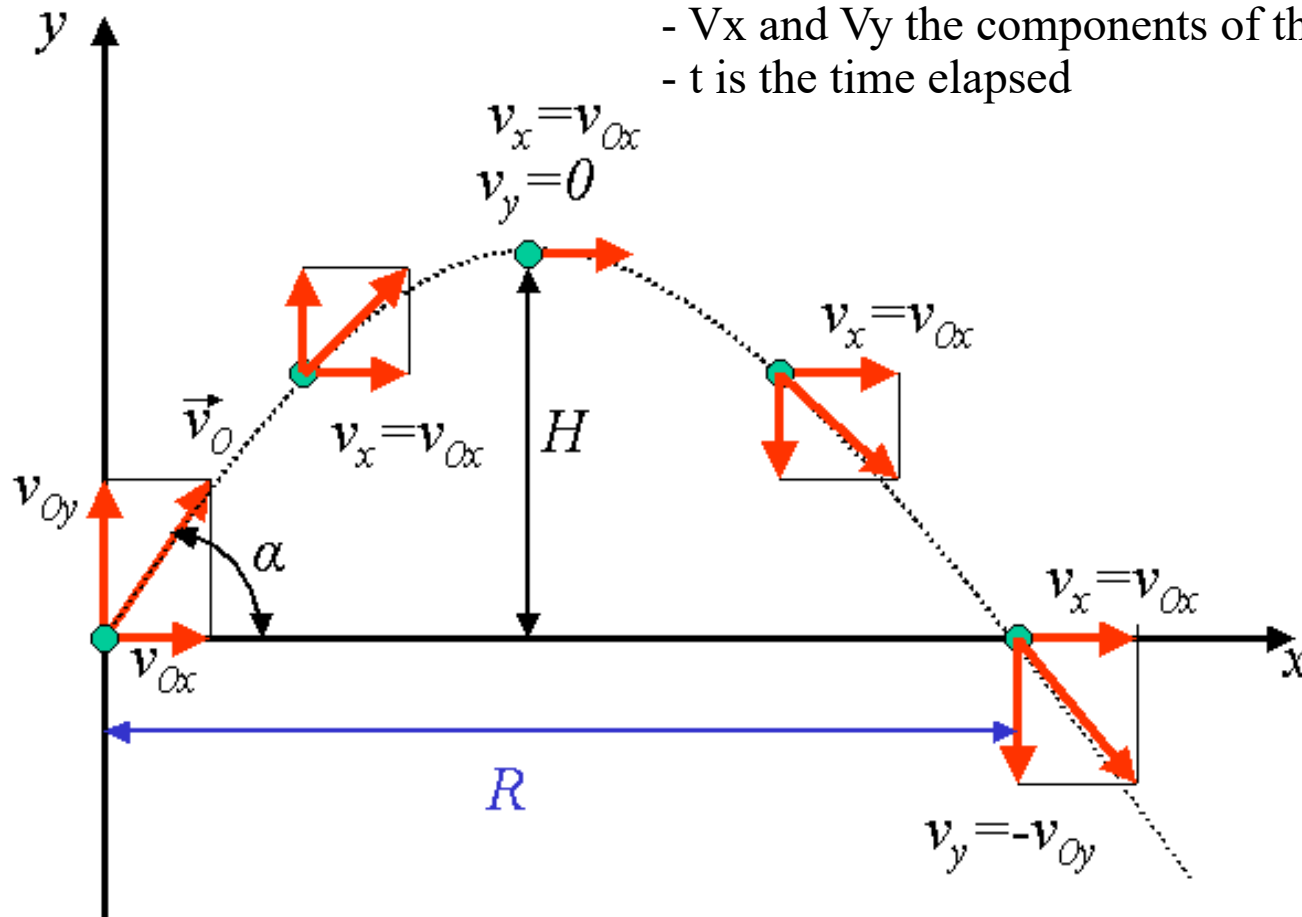
Velocity stays constant.

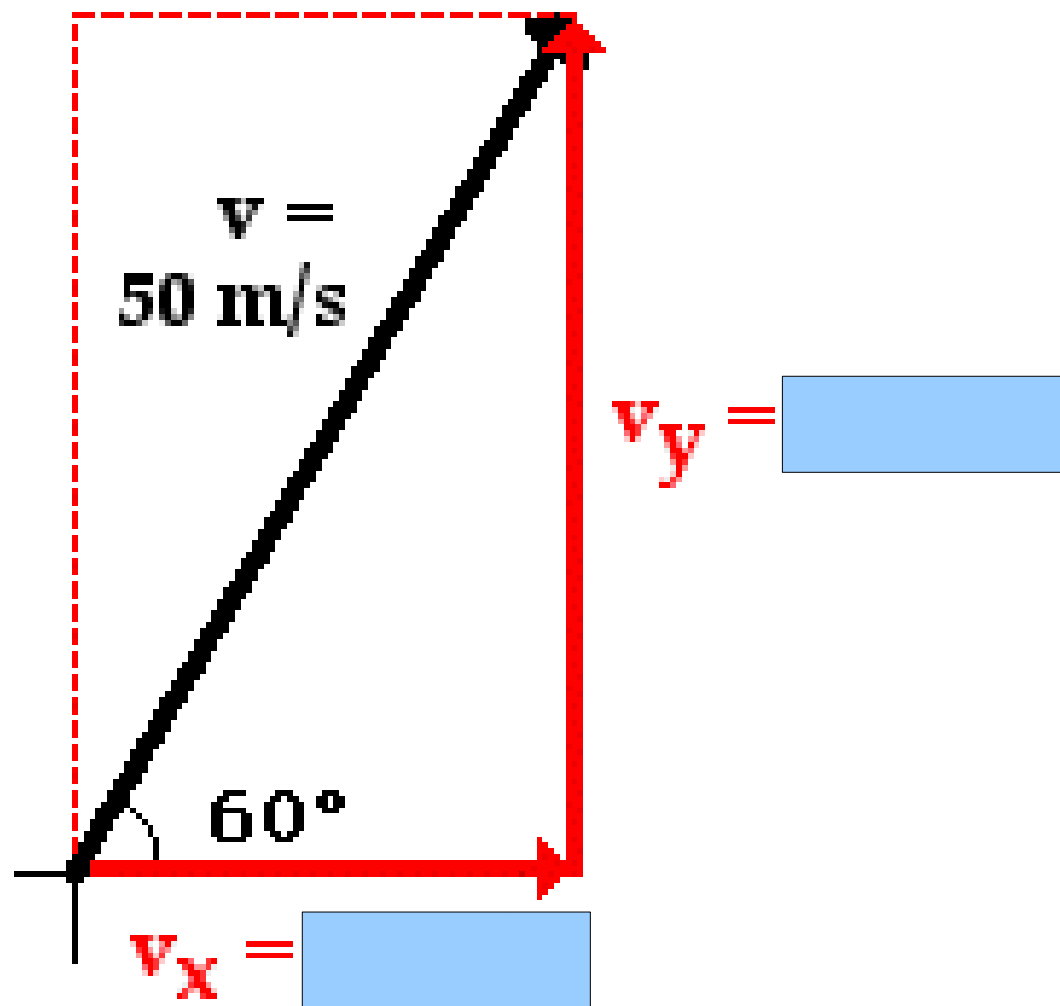
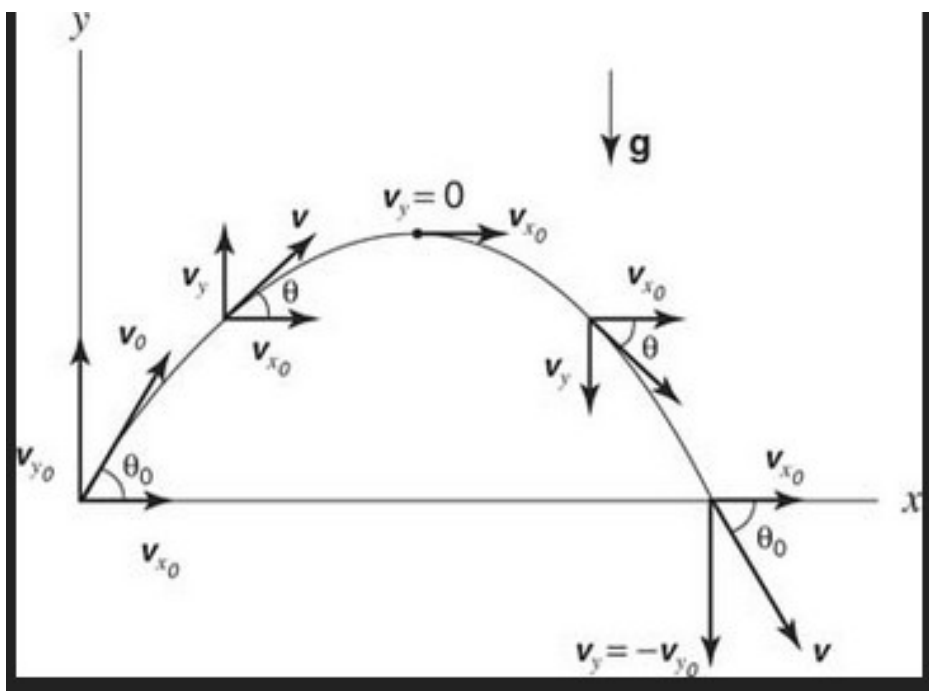
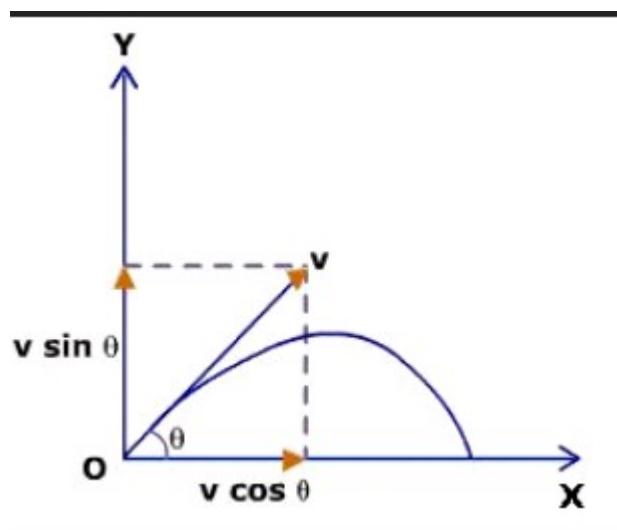
Time t is the same !

a	y	V_{iy}	V_y	t
-10				

Here are the physical quantities that describe Projectile motion:

- x is the horizontal position, y the vertical position
- R is the range of the projectile
- H is the maximum height reached by projectile
- V_{ix} and V_{iy} the components of the initial velocity V_0
- V_x and V_y the components of the velocity along the path
- t is the time elapsed



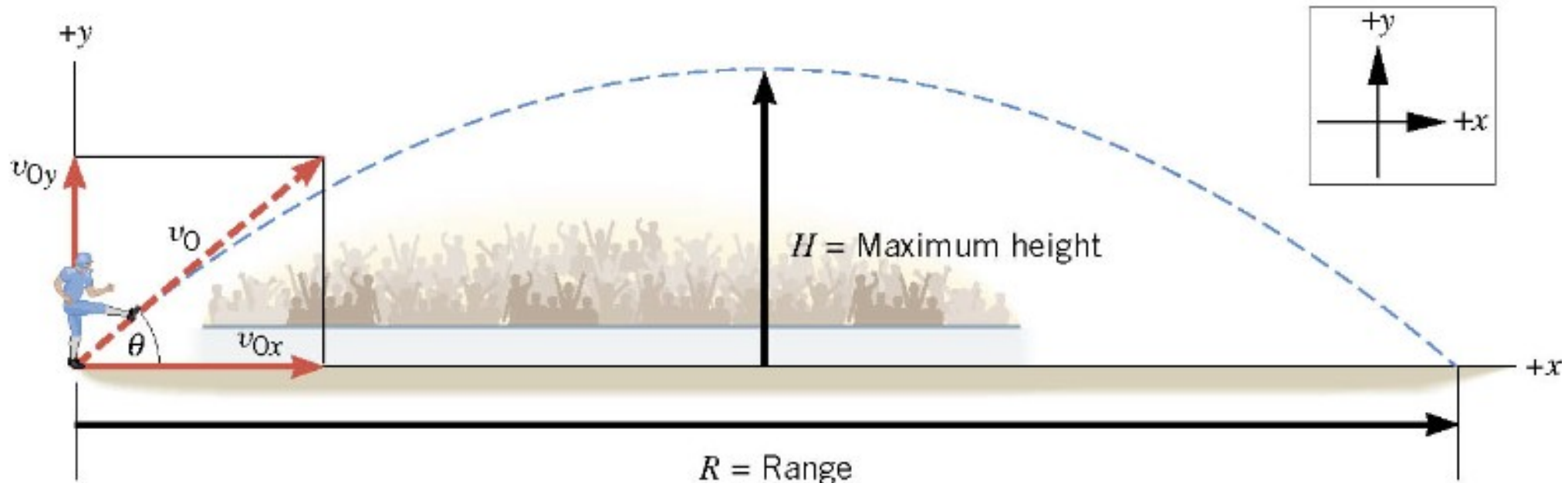


3.3 Projectile Motion

Example 6 The Height of a Kickoff

A placekicker kicks a football at an angle of 40.0° and the initial speed of the ball is $V_0 = 22 \text{ m/s}$. Ignoring air resistance

- 1) Along x: $V_{ix} = ?$ and $V_{iy} = ?$
- 2) the time of flight
- 3) Use the X-axis equation (constant velocity) to find the range
- 4) consider the time when the ball hits the ground (use the time of flight)
Find the components of the final velocity V_x and V_y at that time.
- 5) Find the vector final velocity V given V_x and V_y . (magnitude and direction)
- 6) what's the max height



Along vertical (Y-axis)

free-fall

Acceleration is -10m/s^2

Δy is the vertical displacement (y-axis)

We suppose initial position is 0.

V_{oy} is the initial velocity along y

(V_{oy} is the y-component of V_o the initial velocity)

$$V_o = 22 \sin(40) = 14\text{m/s}$$

$$\Delta Y = V_o t - 5t^2$$

$$V_y = V_o - 10t$$

$$V_y^2 = V_o^2 - 20\Delta Y$$

a	y	V_o	V_y	t
-10				

1) How to find t_{up} ? Set $V=0$ @ top parabola

$$V = 14 - 10t \rightarrow t = 1.4\text{s}$$

$$t_{up} = t_{down} = 1.4\text{s} \text{ so } t_{flight} = 2.8\text{s}$$

2) How to find max height?

$$\Delta Y = V_o t_{up} - 5t_{up}^2 = 10\text{m}$$

Along horizontal no acceleration

$a = 0$

Δx is the horizontal displacement (x-axis)

We suppose initial position is 0.

V_{ox} is the initial velocity along x

(V_{ox} is the x-component of V_o the initial velocity)

$$V_o = 22 \cos(40) = 17\text{m/s}$$

$$\Delta x = V_o t$$

$$V_x = V_o$$

Velocity stays constant.

Time t is the same !

3) How to find the range R?

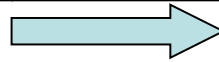
$$R = V_o t_{flight} = 47\text{m}$$

4) final speed = initial speed

Final velocity V (17, -14)

Maximum height $H \rightarrow V_y=0$ at the top of the parabola

y	a_y	v_y	v_{oy}	t
?	-9.80 m/s ²	0	14 m/s	



$$v_y^2 = v_{oy}^2 + 2a_y y$$

$$y = \frac{v_y^2 - v_{oy}^2}{2a_y}$$

$$y = \frac{0 - (14 \text{ m/s})^2}{2(-9.8 \text{ m/s}^2)} = +10 \text{ m}$$

Help Peter

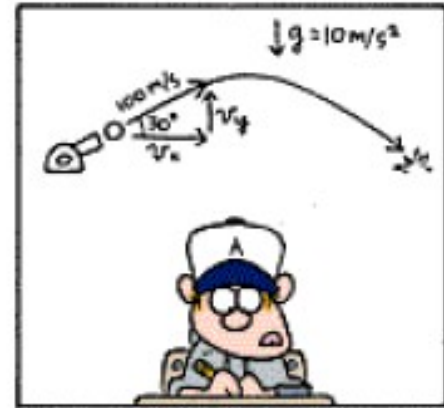
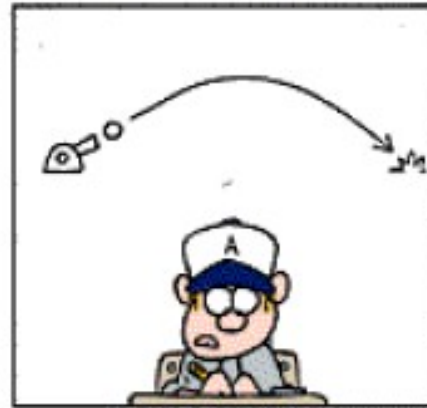
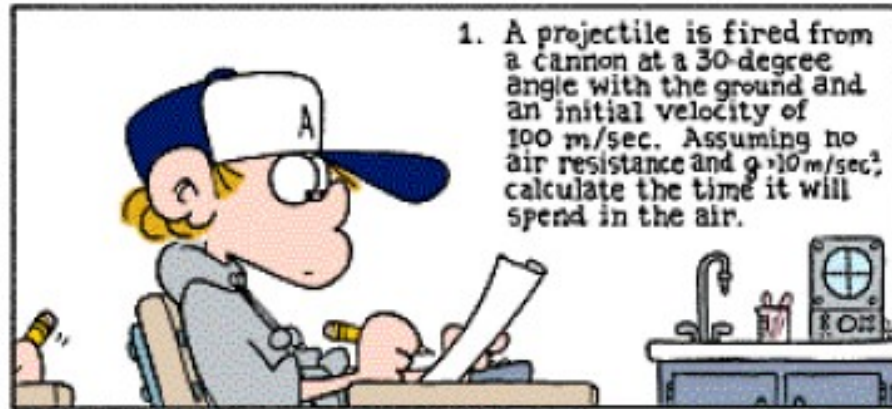
FoxTrot

BILL AMEND

Name: Peter Fox



Date: Not as often as I'd like to, sadly.



5) A ball rolls off a platform that is 5m above the ground.

The ball horizontal velocity as it leaves the platform is $V_{ix} = 6\text{m/s}$.

A) How much time does it take for the ball to hit the ground?

B) How far from the base of the platform does the ball hit the floor? (range or x)

6) A projectile is fired at an angle of 45 degrees. The initial velocity is 40m/s

$V_{ix} = 40\cos(45)$ and $V_{iy} = 40\sin(45)$

A) Using $g = -10$, how long does it take the ball to reach its high point?

B) What horizontal distance does the ball travel in this time?

7)

A cannon is fired over level ground at an angle of 30 degrees to the horizontal.

The initial velocity of the cannon ball is 400m/s. $V_{ix} = 400\cos(30)$ and $V_{iy} = 400\sin(30)$

A) How long is the cannonball in the air? ($g = -10$). Use the y-components.

Remember the total time of flight is twice the time required to reach the high point.

B) How far does the cannonball travel horizontally? Use the x-component.

C) Repeat these calculations assuming the cannon is fired at a 60 degrees angle to the horizontal.

In this case the horizontal component is $400\cos(60)$ and the vertical component is $400\sin(60)$.

D) How does the distance traveled compare to the earlier result?

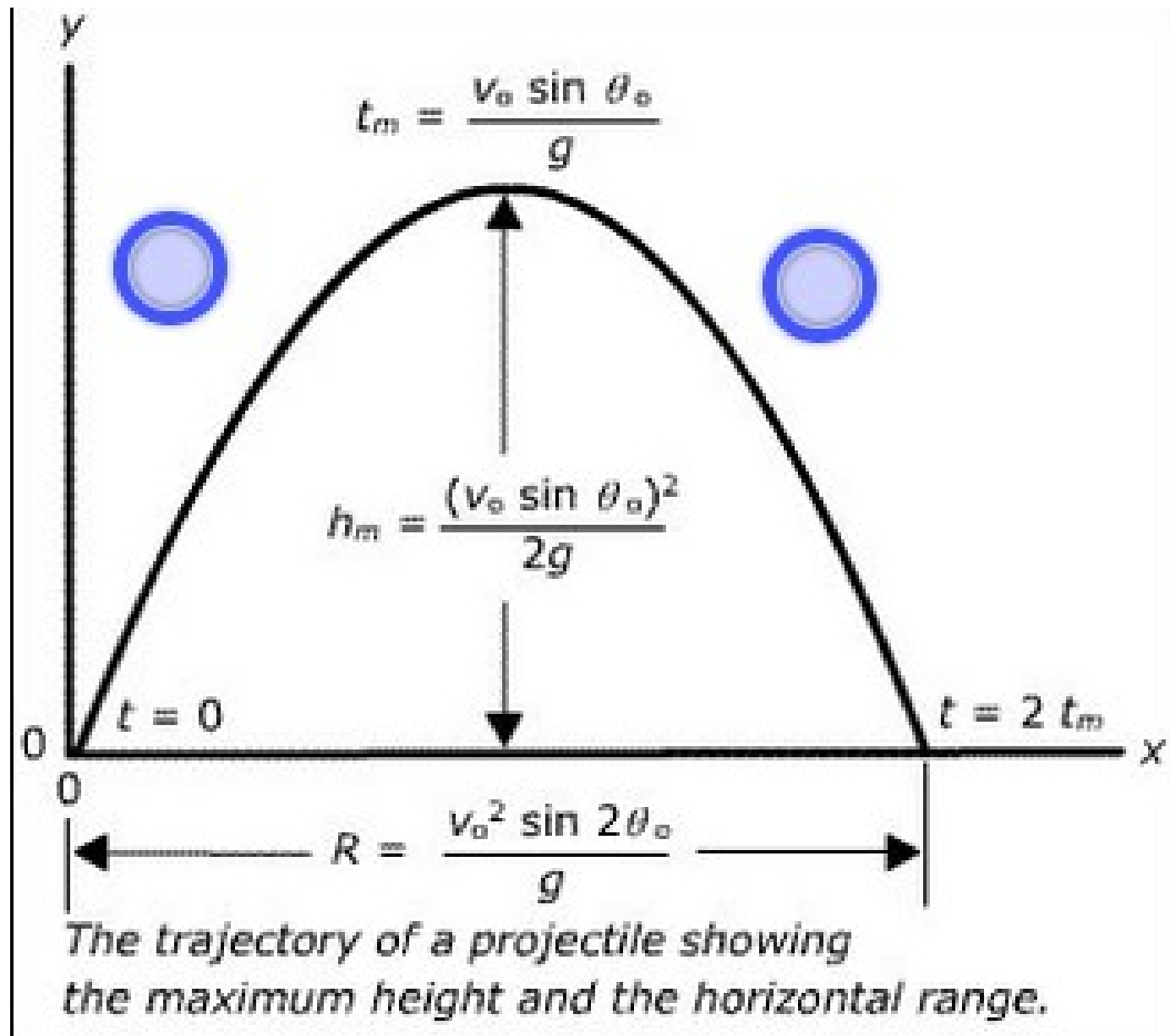
8) 2 balls are rolled off a tabletop that is 0.8m above the floor.

Ball A has a horizontal velocity of $V_{ix} = 3\text{m/s}$ and that of ball B is $V_{ix} = 5\text{m/s}$.

A) Assuming $g = -10$, how long does it take each ball to reach the floor?

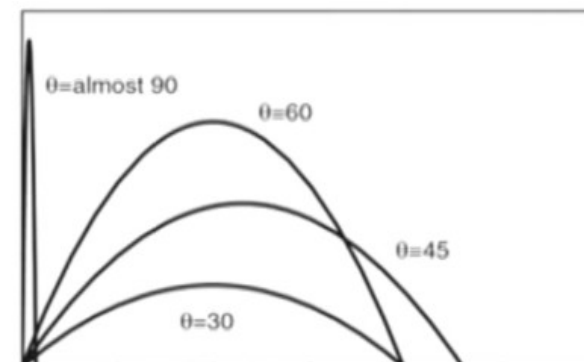
B) How far does each ball travel horizontally before hitting the floor?

Assignments = take home quizzes to do





Jess Ennis

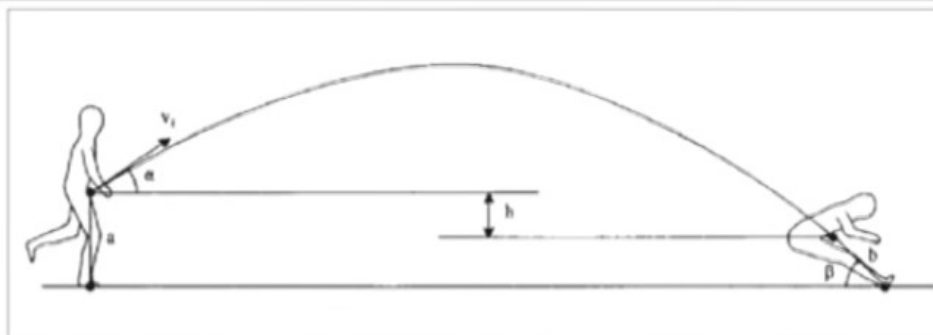


angle θ , the projectile distance can be calculated as follows:

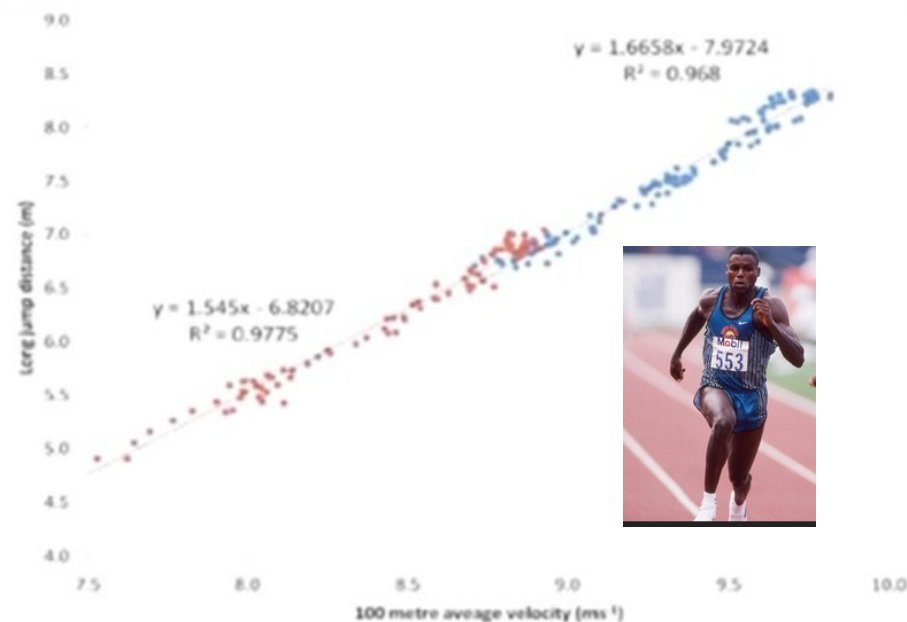
$$\text{Distance} \rightarrow d = \frac{v^2}{g} \sin 2\theta$$

Launch velocity $\rightarrow v^2$
 Launch angle $\rightarrow \theta$
 Acceleration due to gravity $\rightarrow g$

Long jump physics



- Optimum take off angle is less than 45 degrees as the athlete's take off and landing height are different.
- Athletes find it impossible to achieve steep (40 degree +) take off angles.
- Take off angle is typically 33 degrees.



Good runners $\rightarrow$ good long jumper

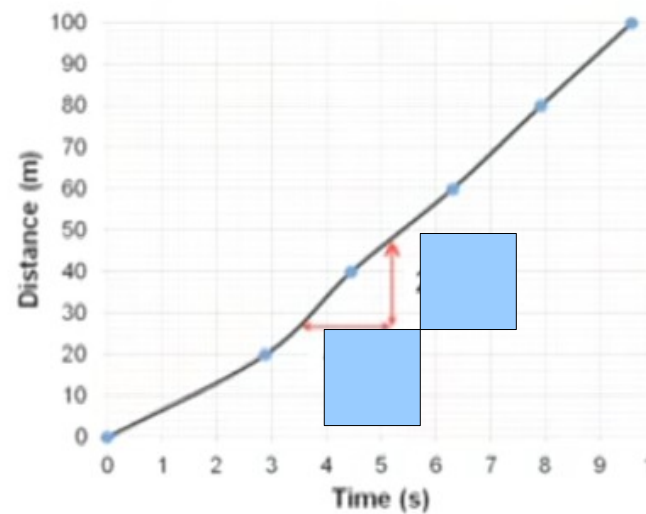
How far could Usain Bolt Jump?



How fast is Usain Bolt?

2009 world record split times :

Distance (m)	0	20	40	60	80	100
Time (s)	0	2.89	4.46	6.31	7.50	9.58



- Gradient of graph is the speed



•At 40 metres, his max sprint speed is 12.1 m/s.

•Assume a 15% loss of speed at take off...

$v = 10.3 \text{ m/s}$

•Bolt's optimum take off angle is 33 degrees.

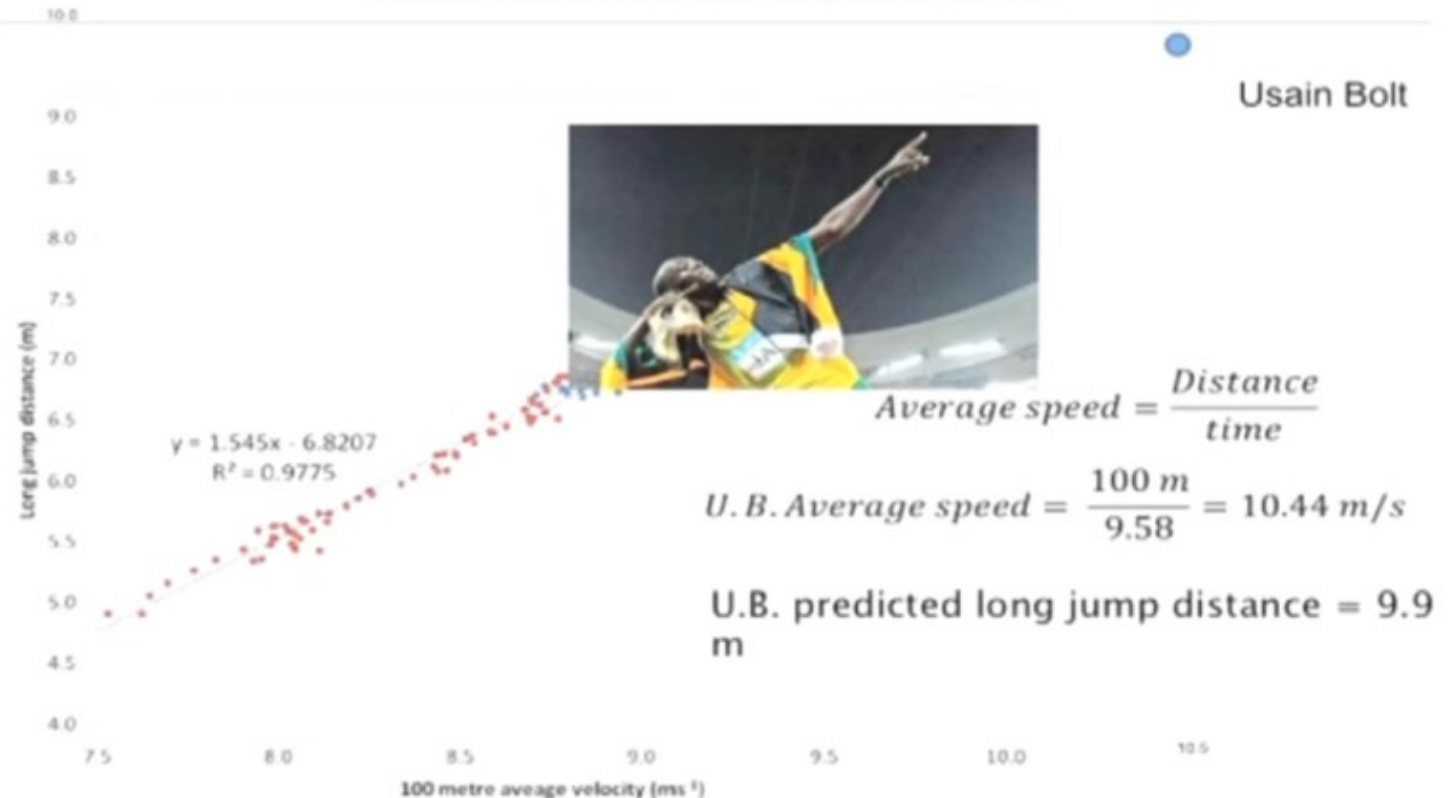
• $\theta = 33^\circ$

•Assuming perfect technique, the theoretical maximum jump distance is...

$$d = \frac{v^2}{g} \sin 2\theta = \frac{\boxed{}^2}{9.8} \sin(2 \times \boxed{}^\circ) = \boxed{} \text{ m}$$



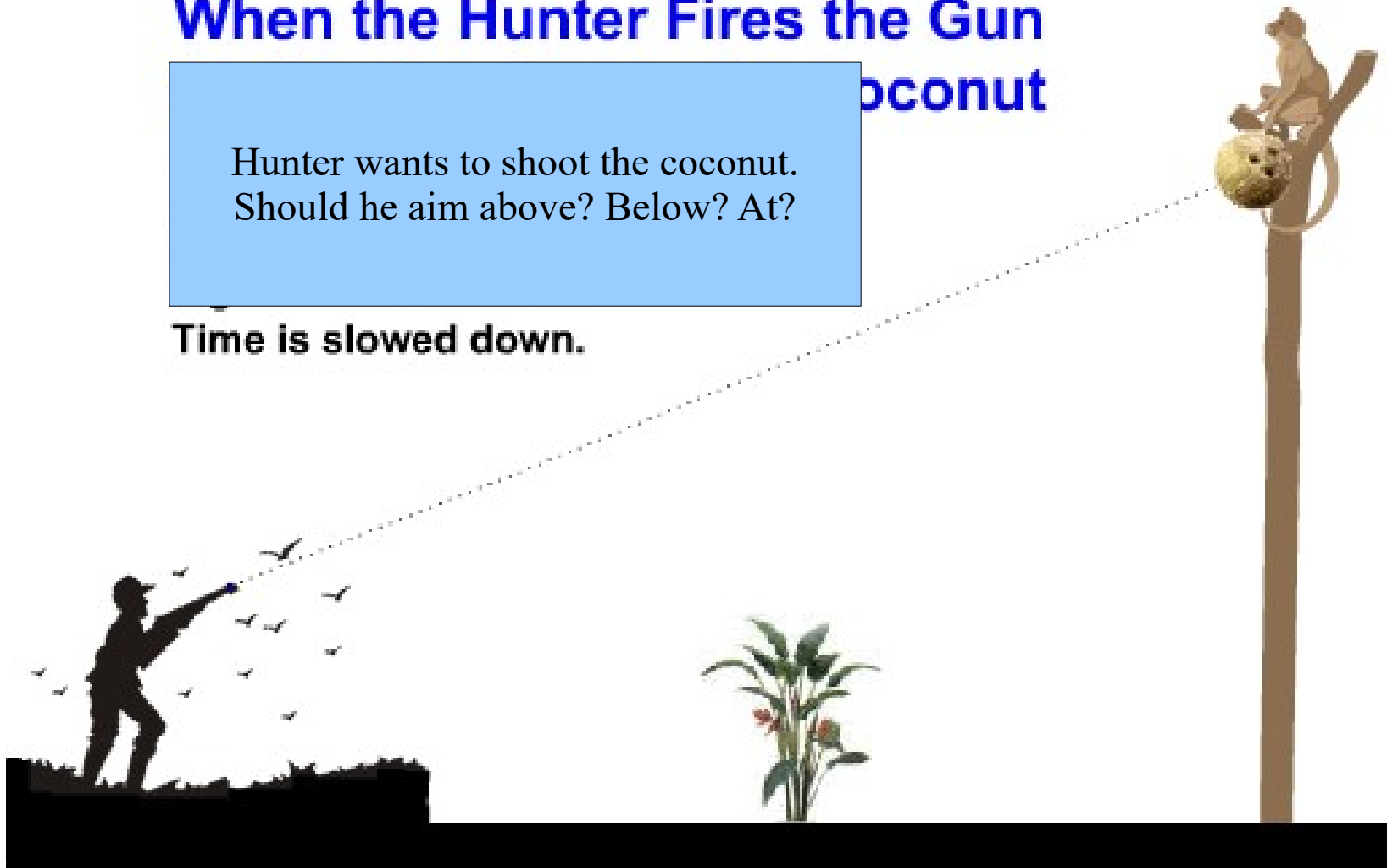
Where would Usain Be?



When the Hunter Fires the Gun Coconut

Hunter wants to shoot the coconut.
Should he aim above? Below? At?

Time is slowed down.

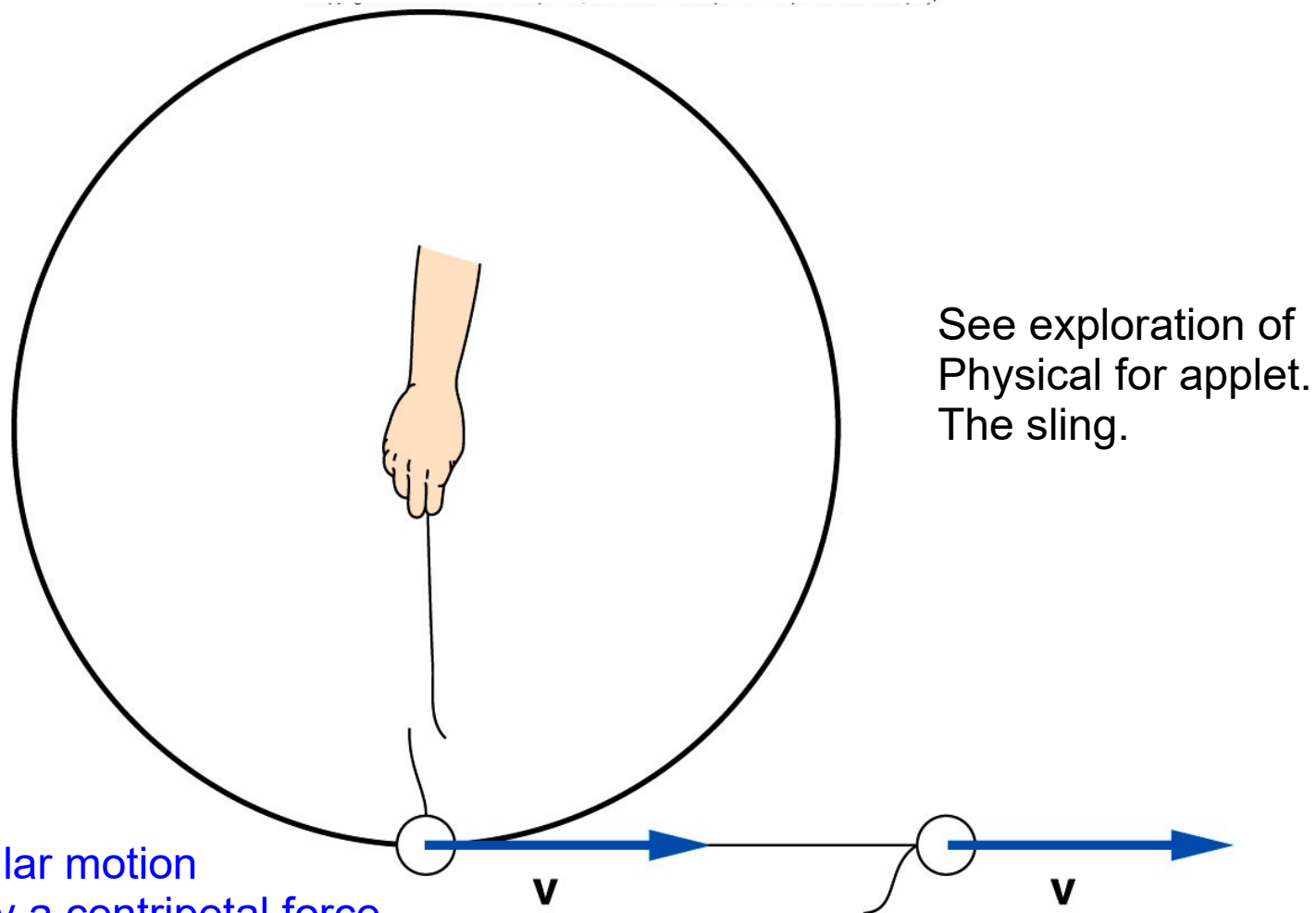


Copyright © 2004 David M. Harrison

Click to begin: 

See exploration software # 7 and video monkey @ MIT

**You spin a ball attached to a string. Which way will it go if you let go ?
How this related to inertia ? And what if the Sun was to disappear ,
which way Will the Earth go ?**



See 2 video circular motion
You need to apply a centripetal force
To keep the object on the track
(demo: glass of water and the broom)

Source: the physics of every day phenomena / Mc Graw Hill

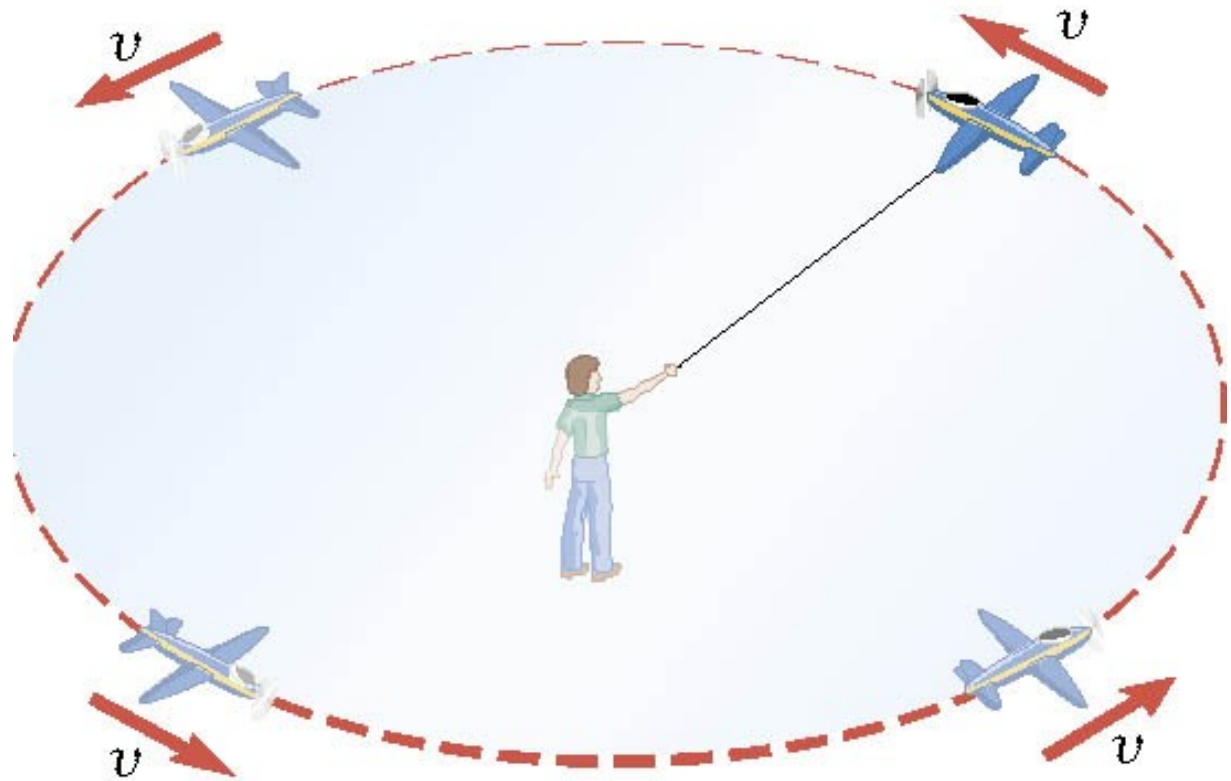
DEFINITION OF UNIFORM CIRCULAR MOTION

Uniform circular motion is the motion of an object traveling at a constant speed on a circular path.

The velocity is a vector.

**The speed is constant
But not the velocity !!!
Can you see why ?**

**The velocity changes in direction.
You need a force to keep the
Plane on the track.
So there is an acceleration !
Force = acceleration !**



For an object in uniform circular motion, which one of the following statements is false ?

- a) The velocity of the object is constant.
- b) The magnitude of the acceleration of the object is constant.
- c) The acceleration is directed radially inward.
- d) The magnitude of the velocity is constant.
- e) The velocity is directed in a direction that is tangent to the circular path.

Video circular motion (which path)

When using the term “uniform circular motion,” what do we mean by the term “uniform?”

- a) The direction of the object’s velocity is constant.
- b) The net force on the moving object is zero newtons.
- c) The forces acting on the object are uniformly applied from all directions.
- d) The motion occurs without the influence of the gravitational force.
- e) The motion of the object is at a constant speed.

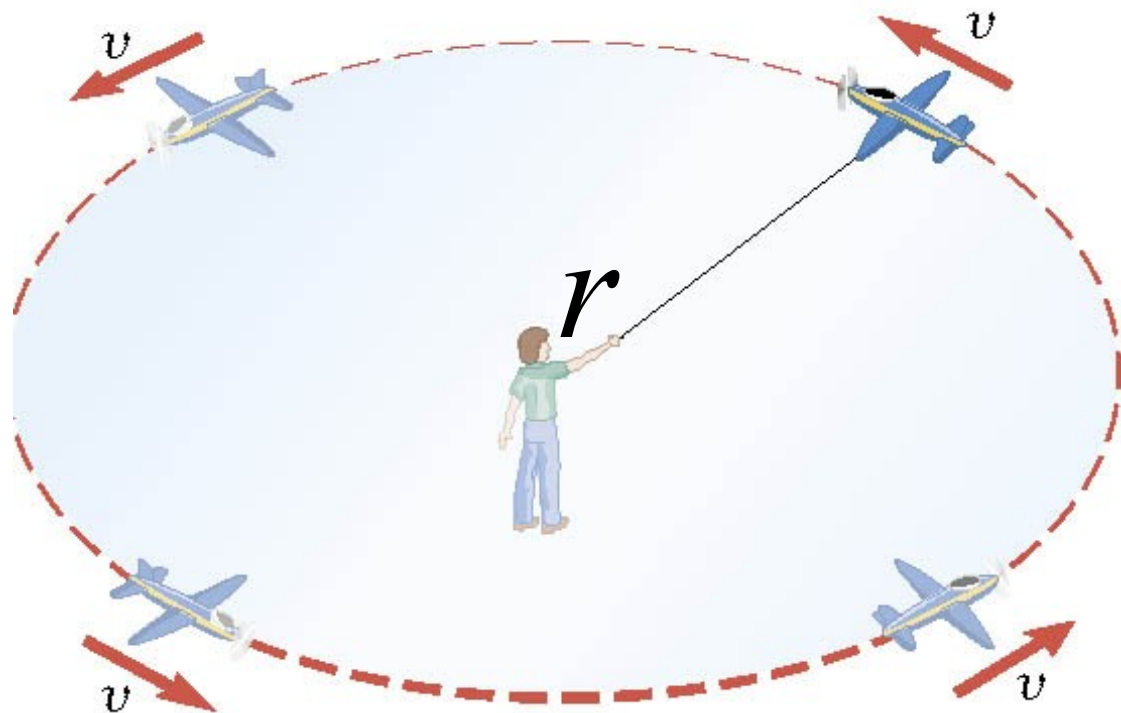
Let T be the time it takes for the object to travel once around the circle.
 T is the period (s). The distance around a circle = $2 \pi r$
(or circumference – 360 degrees)

Note: The distance around half a circle = πr (180 degrees)

The distance around $\frac{1}{4}$ a circle = $2 \pi r / 4$ (90 degrees)

V is a velocity and its direction changes
Its magnitude is constant and is noted v
Here (lower case v in script)

Speed = distance / time
Speed = $2 \pi R$ / time



An airplane flying at 115 m/s due east makes a gradual turn while maintaining its speed and follows a circular path to fly south. The turn takes 15 seconds to complete. What is the radius of the circular path?

a) 410 m

b) 830 m

c) 1100 m

d) 1600 m

e) 1098 m

Round to the nearest hundred.

**From east to south that's $\frac{1}{4}$ of a circular path
(circumference of a circle is $2\pi R$)**

Example

The wheel of a car has a radius of 0.29m and it being rotated at 830 revolutions per minute on a tire-balancing machine. Determine the speed at which the outer edge of the wheel is moving.



REMEMBER ACCELERATION = RATE of CHANGE of velocity

VELOCITY is a vector. VELOCITY CHANGES IF :

- magnitude changes (speed changes) or if
- direction changes !!!

Or BOTH

NOTE: **ACCELERATION IS PUSH/PULL per unit mass.**
ACCELERATION and FORCE have the same DIRECTION !!!

Acceleration is force per unit mass.

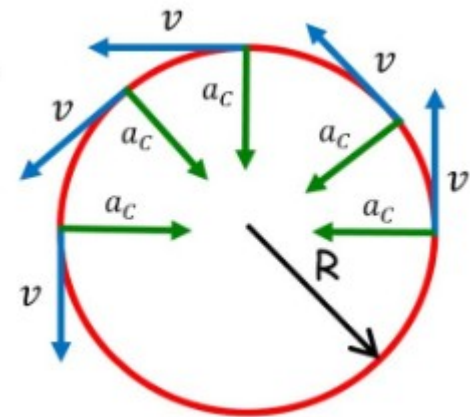
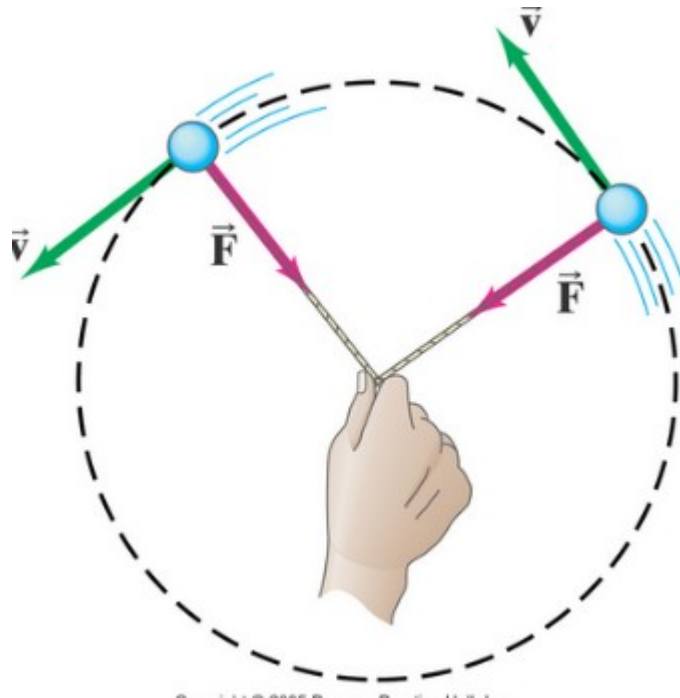
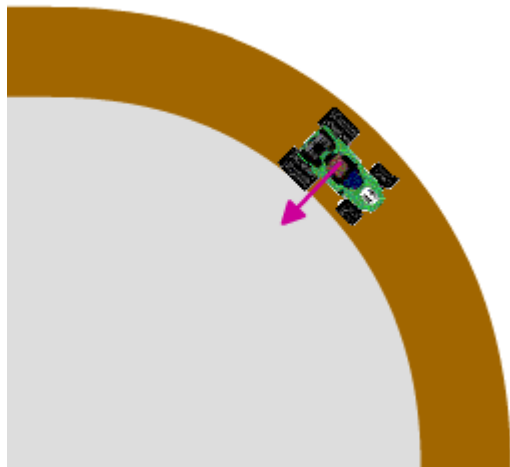
Centripetal acceleration

In uniform circular motion, the speed is constant, but the direction of the velocity vector is *not constant*.

Velocity is perpendicular to the path of the object.

Acceleration is perpendicular to the velocity

$$a_c = \frac{v^2}{r}$$



The direction of the centripetal acceleration is towards the center of the circle; in the same direction as the change in velocity.

A satellite orbits the Earth in uniform circular motion. What is the direction of centripetal acceleration of the satellite?

- a) Because the centripetal acceleration is a scalar quantity, it doesn't have a direction.
- b) The centripetal acceleration vector points radially outward from the Earth.
- c) The centripetal acceleration vector points radially inward toward the Earth.
- d) The centripetal acceleration vector points in the direction of the satellite's velocity.
- e) The centripetal acceleration vector points in the direction opposite that of the satellite's velocity.

Circular Motion is Ubiquitous



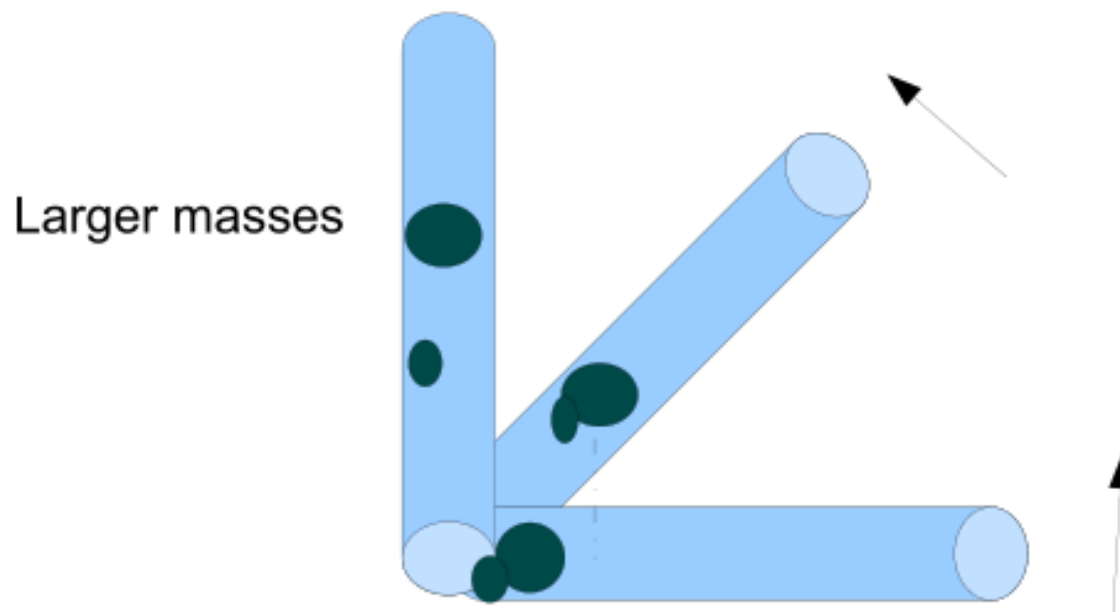
Watch ground
to see turn

SEE MOVIE

A CENTRIFUGE ALSO USES INERTIA TO SEPARATE SUBSTANCES OF DIFFERENT MASSES. A centrifuge can be used to enrich uranium.

Higher technology and hard to make than the calutron.

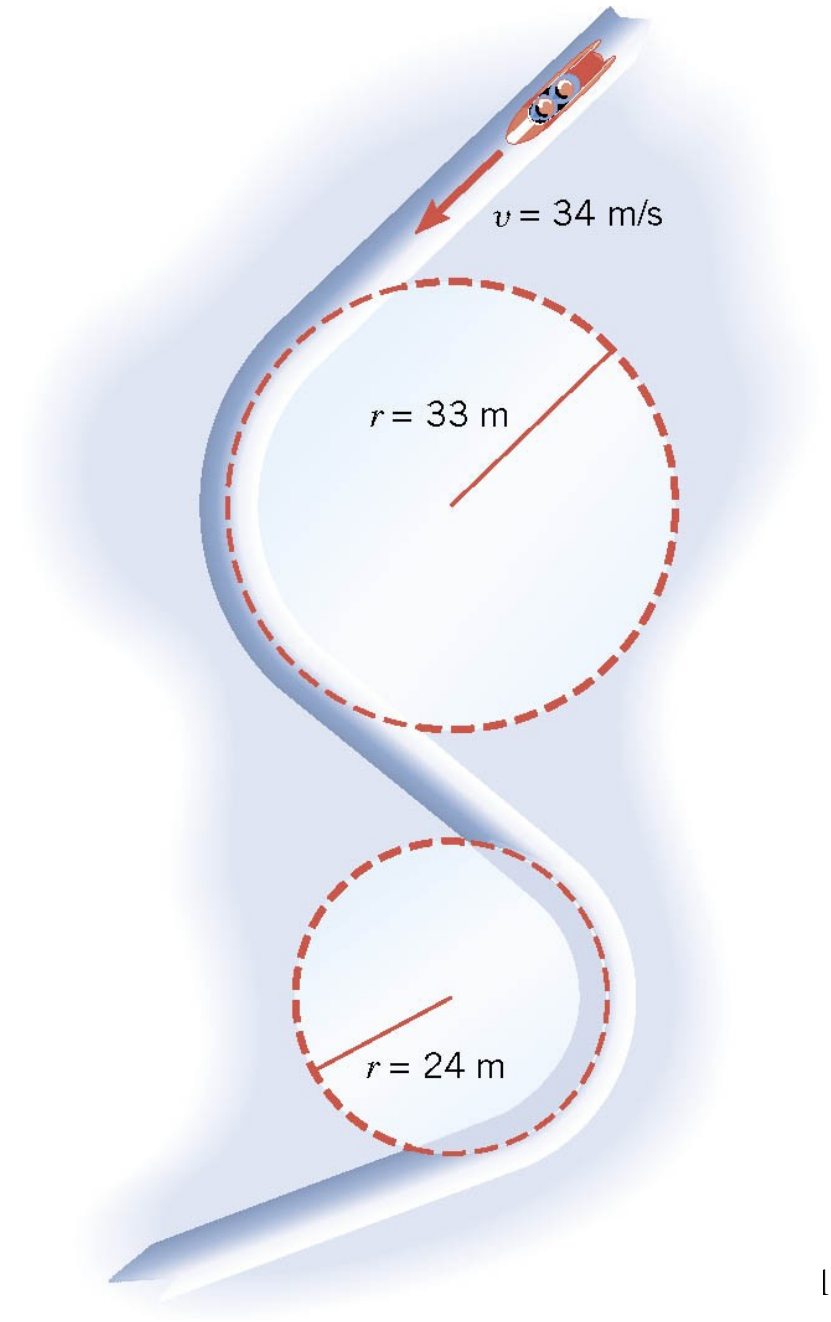
uranium has to be gaseous and combined with fluorine (very corrosive). It is called a gaseous centrifuge http://en.wikipedia.org/wiki/Gas_centrifuge



5.2 Centripetal Acceleration

Example : The Effect of Radius on Centripetal Acceleration

The bobsled track contains turns with radii of 33 m and 24 m. Find the centripetal acceleration at each turn for a speed of 34 m/s. Express answers as multiples of $g = 9.8 \text{ m/s}^2$.



A truck is traveling with a constant speed of 15 m/s. When the truck follows a curve in the road, its centripetal acceleration is 4.0 m/s^2 . What is the radius of the curve?

- a) 3.8 m
- b) 14 m
- c) 56 m
- d) 120 m
- e) 210 m

$$a_c = \frac{v^2}{r} \quad v = \frac{2\pi r}{T}$$

$$a_c = \frac{4\pi^2 r}{T^2}$$

While we are in this classroom, the Earth is orbiting the Sun in an orbit that is nearly circular with an average radius of 1.50×10^{11} m. Assuming that the Earth is in uniform circular motion, what is the centripetal acceleration of the Earth in its orbit around the Sun?

$$a_c = \frac{4\pi^2 r}{T^2}$$

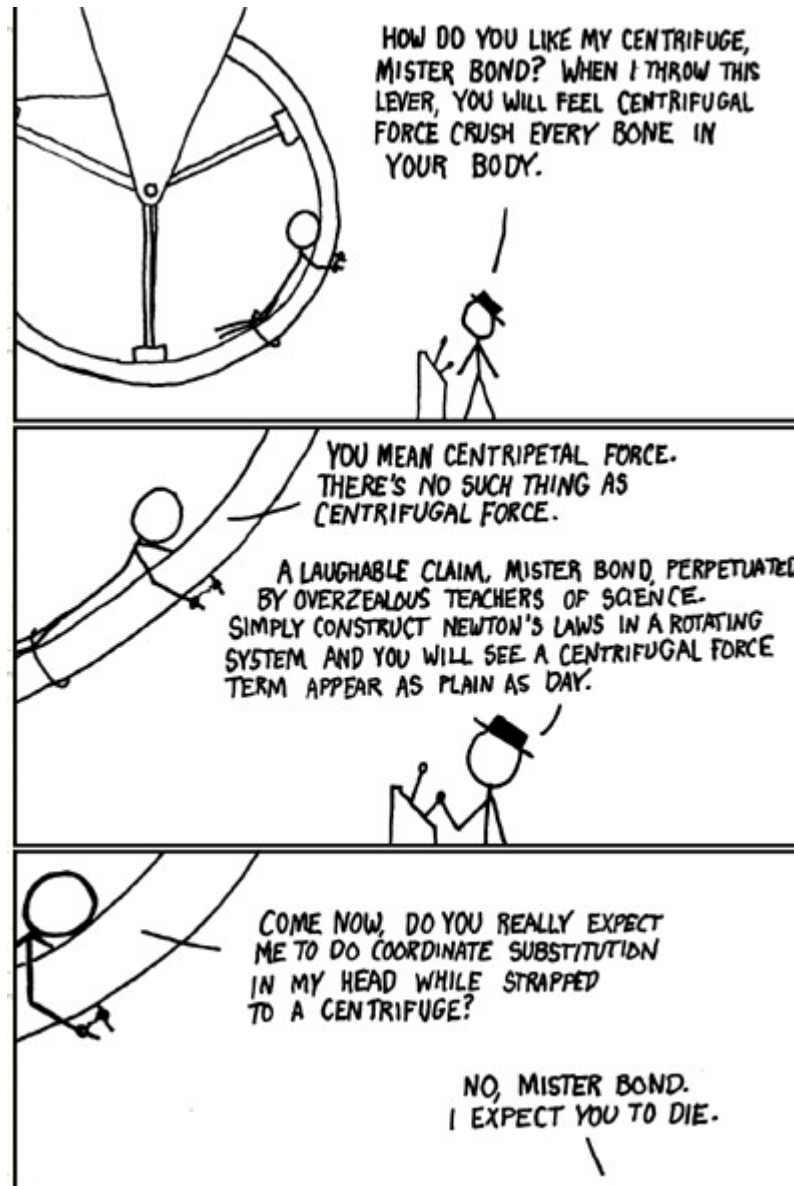
a) $5.9 \times 10^{-3} \text{ m/s}^2$

b) $1.9 \times 10^{-5} \text{ m/s}^2$

c) $3.2 \times 10^{-7} \text{ m/s}^2$

d) $7.0 \times 10^{-2} \text{ m/s}^2$

e) 9.8 m/s^2



source: <http://xkcd.com/556/>

Max acceleration we can support
Is 10gs.

Space shuttle is 3gs → 3 times
Your Weight crushing you.

- 1) What is the acceleration of a motorcycle going 28m/s on a circular track whose radius is 140m ?
- 2) The clothes in a rotary drier whose radius is 0.25m are accelerated toward the center of the drier at 22m/s/s . How fast are the clothes moving ?
- 3) A ball is traveling at a constant speed of 5m/s in a circle with a radius of 0.8m . What is the centripetal acceleration of the ball?
- 4) A car rounds a curve with a radius of 25m at a speed of 20m/s . What is the centripetal acceleration of the car ?
- 5) A ball traveling in a circle with a constant speed of 3m/s has a centripetal acceleration of 9m/s/s . What is the radius of the circle.
- 6) How much larger is the required centripetal acceleration for a car rounding a curve at 60PMH then for one rounding the same curve at 30 MPH ?
- 7) A car with a mass of 1200kg is moving around a curve with a radius of 50m at a constant speed of 20m/s (45MPH)
What is the centripetal acceleration of the car?

optional

8) Object in orbit around the earth (artificial satellites, labs, the Moon..) are in free-fall. A satellite at low orbit has an acceleration of $-10\text{m/s/s} = g$. This acceleration is the centripetal acceleration of the satellite. So $|g|=v^2 / r$ with r = radius of Earth + altitude.

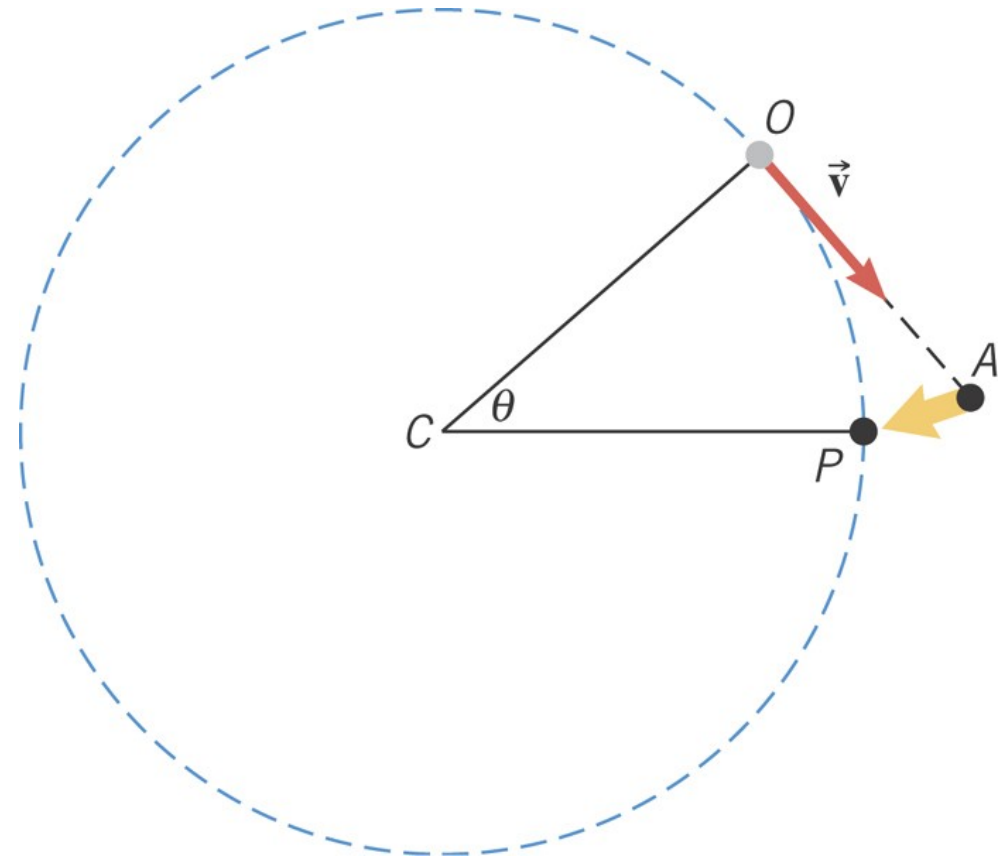
A) An artificial satellite is to be put in orbit a short distance above the surface of the Earth ($r=6,400,000$ m) . What speed must it be given
To make it go into a circular orbit ?

B) What is the acceleration due to gravity g ($|g|=v^2 / r$) at the surface of a planet with a radius of $1,600,000$ m? If a satellite near the
Surface makes one circular orbit every 1700s ? (hint: $v = \text{speed} = 2\pi r / \text{time}$)

C) How fast was the lunar orbiter traveling when it was in orbit around the moon, near its surface ? The radius of the Moon is $3,500,000$ m,
And the acceleration due to gravity at its surface is 1.67m/s/s

Conceptual Example 2: Which Way Will the Object Go?

An object is in uniform circular motion. At point O it is released from its circular path. Does the object move along the straight path between O and A or along the circular arc between points O and P ?



Hint: to change a motion
You need to apply a force.

Changing a motion means: changing direction,
Speed or both.

Video: circular motion: which path

A bicycle racer is traveling at constant speed v around a circular track.

The centripetal acceleration of the bicycle is a_c . What happens to the centripetal acceleration of the bicycle if the speed is doubled to $2v$?

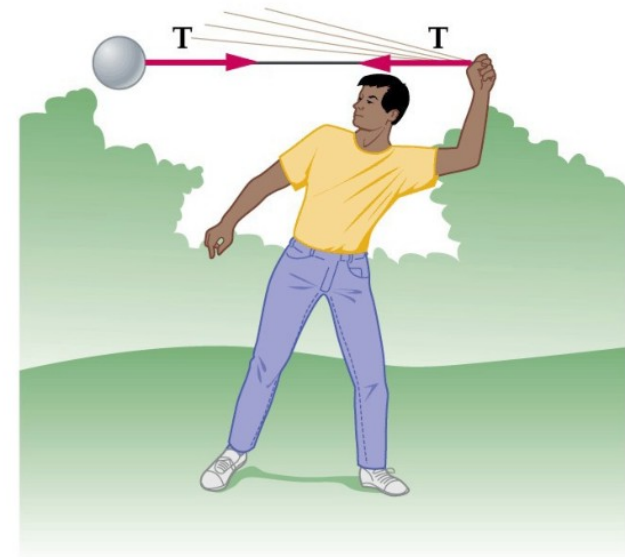
- a) The centripetal acceleration increases to $4a_c$.
- b) The centripetal acceleration decreases to $0.25 a_c$.
- c) The centripetal acceleration increases to $2a_c$.
- d) The centripetal acceleration decreases to $0.5a_c$.
- e) The centripetal acceleration does not change.

5.7.3. A steel ball is tied to the end of a string and swung in a vertical circle at constant speed. Complete the following statement: The direction of the instantaneous velocity of the ball is always

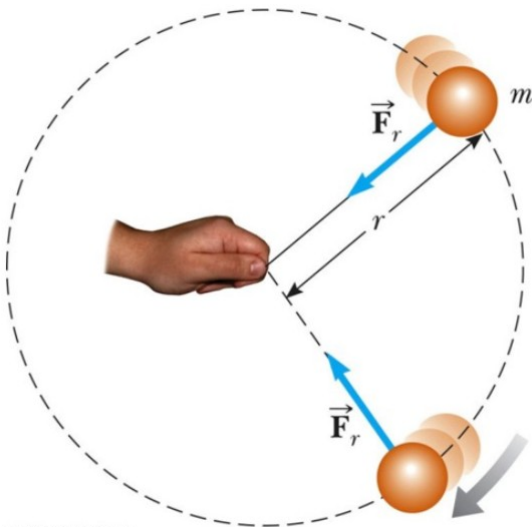
- a) perpendicular to the circle.
- b) toward the center of the circle.
- c) tangent to the circle.
- d) radially outward from the circle.
- e) vertically downward.

Circular Motion: Observations

- Object moving along a curved path with **constant speed**
 - Magnitude of velocity: same
 - Direction of velocity: changing
 - Velocity: changing
 - Acceleration is NOT zero!
 - **Net force acting on the object is NOT zero**
 - "Centripetal force"



$$\vec{F}_{net} = m \vec{a}$$



$$V = 2\pi R / T$$

For the same $T \rightarrow V$ is proportional to R

<https://www.flippingphysics.com/rotational-kinematics-gifs.html>